Python Practice Workbook

Module 5: Integrated Projects

End-to-End Time Series Analysis

Time Series Analytics T6725 — M.Sc. Data Science Sem IV

Symbiosis International (Deemed) University

Datasets: BTC-USD — MRF.NS — GC=F

Standard Setup Block — Run this first in every Colab session

```
!pip install yfinance statsmodels arch pmdarima -q
import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
```

Module 5 Design Philosophy: Every problem here is **integrative** — it requires knowledge from Modules 1 through 4 simultaneously. Every problem asks students to: (1) fetch real data, (2) diagnose it, (3) select and fit the right model, (4) validate it, (5) forecast it, and (6) interpret the result for a real stakeholder.

Problem Distribution

Difficulty	Count	Format	Primary Dataset
EASY	5	Guided mini-projects with scaffolding	All three
MEDIUM	10	Partial scaffolding, student completes	BTC & MRF
HARD	13	Open-ended with target deliverables	All three
EXAM	12	Full capstone projects, multi-part	All three
TOTAL	**40**	All 8 types	All three

Section A Easy Guided Mini-Projects — M5.001 to M5.005

M5.001 *Guided Mini-Project* **EASY**

Topic: BTC Complete EDA in 30 Lines

The Brief: You just joined a crypto analytics startup. Your first task: produce a quick 4-panel exploratory report on Bitcoin returns so the team can decide whether ARIMA or GARCH is the right next step.

Fill in the **eight blanks** to complete the pipeline.

```
btc     = yf.download("BTC-USD", start="2019-01-01",
   end="2024-12-31")
close   = btc["______"].dropna()
returns = np.log(close).______.dropna() * 100

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle("BTC-USD -- Exploratory Report", fontsize=13,
   fontweight="bold")

# Panel 1: Price series
close.plot(ax=axes[0,0], color="orange")
axes[0,0].set_title("Price (USD)")

# Panel 2: Log returns
returns.plot(ax=axes[______], color="steelblue",
   linewidth=0.6)
axes[0,1].set_title("Log Returns (%)")

# Panel 3: ACF of returns
from statsmodels.graphics.tsaplots import plot_acf
plot_acf(______, lags=30, ax=axes[1,0], zero=False,
   alpha=0.05)
axes[1,0].set_title("ACF of Returns -- MA order?")

# Panel 4: ACF of squared returns
plot_acf(returns_____2, lags=30, ax=axes[1,1],
   zero=False, alpha=0.05)
axes[1,1].set_title("ACF of Squared Returns -- ARCH
   effects?")

plt.tight_layout()
plt.show()

from statsmodels.stats.diagnostic import acorr_ljungbox
from arch.univariate import arch_lm_test

lb_ret = acorr_ljungbox(returns, lags=20, return_df=True)
lm_sq  = arch_lm_test(returns, nlags=10)

print(f"Mean autocorrelation (LB min p):
```

```
    {lb_ret['lb_pvalue'].______():.4f}")
print(f"Variance autocorrelation (LM p):
    {lm_sq.pvalue:.6f}")

if lm_sq.pvalue < 0.05:
    print("-> GARCH required for volatility modelling")
if lb_ret['lb_pvalue'].min() < 0.05:
    print("-> ______ may help for mean modelling")
else:
    print("-> Returns are near white noise -- ARIMA mean
        model limited")
```

Solution

① Correct Code

```
!pip install yfinance statsmodels arch -q
import yfinance as yf, numpy as np
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from statsmodels.graphics.tsaplots import plot_acf
from statsmodels.stats.diagnostic import acorr_ljungbox
from arch.univariate import arch_lm_test

btc     = yf.download("BTC-USD", start="2019-01-01",
    end="2024-12-31")
close   = btc["Close"].dropna()              # column name
    is "Close"
returns = np.log(close).diff().dropna() * 100  # .diff()
    for first difference

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle("BTC-USD -- Exploratory Report", fontsize=13,
    fontweight="bold")

close.plot(ax=axes[0,0], color="orange")
axes[0,0].set_title("Price (USD)")

returns.plot(ax=axes[0,1], color="steelblue",
    linewidth=0.6)  # [0,1] = row 0, col 1
axes[0,1].set_title("Log Returns (%)")

plot_acf(returns, lags=30, ax=axes[1,0], zero=False,
    alpha=0.05)  # raw returns
axes[1,0].set_title("ACF of Returns -- MA order?")

plot_acf(returns**2, lags=30, ax=axes[1,1], zero=False,
    alpha=0.05)  # squared: **2
```

```python
axes[1,1].set_title("ACF of Squared Returns -- ARCH effects?")

plt.tight_layout(); plt.show()

lb_ret = acorr_ljungbox(returns, lags=20, return_df=True)
lm_sq  = arch_lm_test(returns, nlags=10)

print(f"Mean autocorrelation (LB min p)  : {lb_ret['lb_pvalue'].min():.4f}")  # .min()
print(f"Variance autocorrelation (LM p)  : {lm_sq.pvalue:.6f}")

if lm_sq.pvalue < 0.05:
    print("-> GARCH required for volatility modelling")
if lb_ret['lb_pvalue'].min() < 0.05:
    print("-> ARIMA may help for mean modelling")  # ARIMA for mean structure
else:
    print("-> Returns are near white noise -- ARIMA mean model limited")
```

② Plain English

This 4-panel EDA is the universal starting point for any financial time series. Panel 1 confirms the asset's price history. Panel 2 shows the stationarity of returns. Panel 3 diagnoses whether an ARIMA model could improve on white noise. Panel 4 is the critical GARCH diagnostic — significant ACF in squared returns means GARCH is needed. For BTC, Panel 4 will show strong spikes (ARCH confirmed) while Panel 3 will be mostly flat (ARIMA limited).

③ Common Mistake

Plotting the ACF of raw prices (Panel 3 should use `returns`, not `close`). Raw prices have non-stationary ACF that decays slowly — it appears dramatic but is statistically meaningless. Always ACF-test the stationary series (returns), not the levels (prices).

④ Real World Link

This EDA is the "60-second pitch" to a quant team: "BTC returns are near white noise in mean (ACF plot) but have massive variance clustering (squared ACF). This means we need GARCH, not ARIMA, as the primary model." The two test p-values quantify exactly this story.

M5.002 *Guided Mini-Project* **EASY**

Topic: MRF — From Raw Data to ARIMA Forecast in 8 Steps
The Brief: Your professor asks you to demonstrate the complete Box-Jenkins pipeline on MRF in a single notebook cell, with every step labelled.
Fill in the blanks to complete all 8 steps.

```python
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.stats.diagnostic import acorr_ljungbox

# STEP 1: Fetch
mrf    = yf.download("MRF.NS", start="2019-01-01",
   end="2024-12-31")
close = mrf["Close"].______()

# STEP 2: Transform
ret    = np.log(close).______().dropna() * 100

# STEP 3: Stationarity test
p_adf = adfuller(ret)[______]
print(f"Step 3 -- ADF p: {p_adf:.6f} -> {'Stationary' if
   p_adf<0.05 else 'Non-stationary'}")

# STEP 4: ACF/PACF
from statsmodels.graphics.tsaplots import plot_acf,
   plot_pacf
fig, ax = plt.subplots(1,2,figsize=(14,4))
plot_acf( ret, lags=20, ax=ax[0], zero=False);
   ax[0].set_title("ACF -> q?")
plot_pacf(ret, lags=20, ax=ax[1], zero=False);
   ax[1].set_title("PACF -> p?")
plt.tight_layout(); plt.______.____()

# STEP 5: Fit ARIMA(1,0,1)
fit = ARIMA(ret, order=(1,______,1)).fit()

# STEP 6: Check residuals
lb = acorr_ljungbox(fit.______, lags=20, return_df=True)
print(f"Step 6 -- Residual LB min p:
   {lb['lb_pvalue'].min():.4f}")

# STEP 7: AIC
print(f"Step 7 -- Model AIC: {fit.______:.2f}")

# STEP 8: Forecast 10 days
fc = fit.forecast(steps=______)
print(f"Step 8 -- 10-day forecast range: [{fc.min():.4f}%,
   {fc.max():.4f}%]")
```

Solution

① Correct Code

```
!pip install yfinance statsmodels arch -q
```

```
import yfinance as yf, numpy as np
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.stats.diagnostic import acorr_ljungbox
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf

# STEP 1: Fetch
mrf   = yf.download("MRF.NS", start="2019-01-01", end="2024-12-31")
close = mrf["Close"].dropna()                    # .dropna() removes NaN

# STEP 2: Transform
ret   = np.log(close).diff().dropna() * 100      # .diff() for first difference

# STEP 3: Stationarity test
p_adf = adfuller(ret)[1]                         # index [1] = p-value
print(f"Step 3 -- ADF p: {p_adf:.6f} -> {'Stationary' if p_adf<0.05 else 'Non-stationary'}")

# STEP 4: ACF/PACF
fig, ax = plt.subplots(1,2,figsize=(14,4))
plot_acf( ret, lags=20, ax=ax[0], zero=False); ax[0].set_title("ACF -> q?")
plot_pacf(ret, lags=20, ax=ax[1], zero=False); ax[1].set_title("PACF -> p?")
plt.tight_layout(); plt.show()                   # plt.show() to display

# STEP 5: Fit ARIMA(1,0,1) -- d=0 because returns already stationary
fit = ARIMA(ret, order=(1,0,1)).fit()

# STEP 6: Check residuals
lb = acorr_ljungbox(fit.resid, lags=20, return_df=True)  # .resid attribute
print(f"Step 6 -- Residual LB min p: {lb['lb_pvalue'].min():.4f}")

# STEP 7: AIC
print(f"Step 7 -- Model AIC: {fit.aic:.2f}")     # .aic attribute

# STEP 8: Forecast 10 days
```

```
fc = fit.forecast(steps=10)                                    # steps=10
print(f"Step 8 -- 10-day forecast range: [{fc.min():.4f}%, {fc.max():.4f}%]")
```

② Plain English

This 8-step pipeline is the complete Box-Jenkins workflow in its most compact form. Memorising this sequence — Fetch → Transform → Test → Visualise → Fit → Validate → Score → Forecast — gives you a repeatable framework for any time series dataset.

③ Common Mistake

Setting `order=(1,1,1)` in Step 5 when the input is already returns (stationary) — this double-differences the data. If your input is returns, set `d=0`. Only set `d=1` when your input is the raw price level. The ADF test in Step 3 should inform this choice.

④ Real World Link

MRF is a NIFTY 50 constituent. An ARIMA forecast on MRF returns would be used by a systematic trading desk to model the expected daily return and compute position sizes. The 10-day forecast converging toward zero reflects the near-efficient nature of MRF's return process.

M5.003 *Guided Mini-Project* **EASY**

Topic: Gold — Seasonal Decomposition to SARIMA

The Brief: Gold has a documented seasonal demand pattern driven by Indian festivals and Chinese New Year. Confirm this seasonality visually and fit a model that captures it.

Fill in the blanks.

```
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.statespace.sarimax import SARIMAX

gold    = yf.download("GC=F", start="2019-01-01", end="2024-12-31")
monthly = gold["Close"].dropna().resample("______").last()

# Decompose
result  = seasonal_decompose(monthly, model="______", period=______)
result.plot()
plt.suptitle("Gold Monthly -- Multiplicative Decomposition", fontsize=11)
plt.tight_layout(); plt.show()

# First year of seasonal factors
seasonal_factors = result.seasonal[:"______-12"]
print("Seasonal factors (first year):")
print(seasonal_factors.round(4))

```

```
# Fit SARIMA(1,1,1)(1,1,0)[12]
model = SARIMAX(monthly, order=(1,1,1),
                seasonal_order=(1,______,0,______),
                enforce_stationarity=False,
                enforce_invertibility=False)
fit   = model.fit(______)
print(fit.summary())

# 12-month forecast
fc = fit.______(steps=12)
print(f"12-month forecast range: [${fc.min():,.2f},
   ${fc.max():,.2f}]")
```

Solution

① Correct Code

```
!pip install yfinance statsmodels -q
import yfinance as yf, numpy as np
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.statespace.sarimax import SARIMAX

gold    = yf.download("GC=F", start="2019-01-01",
   end="2024-12-31")
monthly = gold["Close"].dropna().resample("ME").last()  #
   "ME" = Month End

result  = seasonal_decompose(monthly,
   model="multiplicative", period=12)
result.plot()
plt.suptitle("Gold Monthly -- Multiplicative
   Decomposition", fontsize=11)
plt.tight_layout(); plt.show()

seasonal_factors = result.seasonal["2019-12"]  # end of
   2019
print("Seasonal factors (first year):");
   print(seasonal_factors.round(4))
print("(>1 = above average demand month, <1 = below
   average)")

# SARIMA(1,1,1)(1,1,0)[12]: D=1 seasonal diff, s=12 months
model = SARIMAX(monthly, order=(1,1,1),
   seasonal_order=(1,1,0,12),
                enforce_stationarity=False,
                   enforce_invertibility=False)
```

```python
fit   = model.fit(disp=False)  # disp=False suppresses iteration output
print(fit.summary())

fc = fit.forecast(steps=12)  # .forecast() method
print(f"12-month forecast range: [${fc.min():,.2f}, ${fc.max():,.2f}]")
```

② Plain English

This pipeline takes Gold from raw download to a seasonal forecast in six steps. The decomposition reveals whether October–December shows higher seasonal factors (Diwali/Christmas buying) vs July–August (seasonal lows). SARIMA then explicitly models these calendar patterns, giving forecasts that rise and fall with the seasonal rhythm.

③ Common Mistake

Using `model="additive"` for Gold. Gold's price more than doubled from 2019 to 2024 — the seasonal swings in absolute dollar terms also doubled. This is multiplicative behaviour. Additive decomposition would assign fixed-dollar seasonal effects calibrated to 2019 prices and apply them unchanged to 2024's much higher price level.

④ Real World Link

Gold importers and jewellers in India use seasonal Gold price forecasts for inventory planning. If the SARIMA model predicts October Gold prices 8% higher than August (Diwali effect), a smart jeweller stocks up in August before the seasonal rally. The model's seasonal coefficients translate directly into business procurement strategy.

M5.004 *Guided Mini-Project* **EASY**

Topic: BTC — GARCH Volatility Report Card

The Brief: A risk manager asks: "Give me a one-page GARCH summary for BTC." Build it.

```python
from arch import arch_model

btc     = yf.download("BTC-USD", start="2019-01-01", end="2024-12-31")
returns = np.log(btc["Close"]).diff().dropna() * 100

fit = arch_model(returns, mean="Constant",
                 vol="______", p=1, q=1,
                    dist="t").fit(disp="off")

omega      = fit.params["______"]
alpha      = fit.params["alpha[1]"]
beta       = fit.params["______"]
nu         = fit.params["nu"]
```

```
alpha_beta = alpha + beta
lrv        = omega / (1 - ______)
lrvol      = np.sqrt(lrv)
hl         = np.log(0.5) / np.log(______)
latest_vol   = fit.conditional_volatility.______[-1]
latest_vol_a = latest_vol * np.sqrt(252)

print("="*50)
print("  BTC-USD GARCH(1,1) -- Risk Report Card")
print("="*50)
print(f"  Persistence (a+b)     : {alpha_beta:.4f}")
print(f"  Shock half-life       : {hl:.1f} trading days")
print(f"  Long-run annual vol   :
   {lrvol*np.sqrt(______):.2f}%")
print(f"  Current annual vol    : {latest_vol_a:.2f}%")
print(f"  t degrees of freedom : {nu:.2f}")
print(f"  Tail classification   : {'Extreme fat tails' if
   nu<5 else 'Heavy tails'}")
print("="*50)
```

Solution

① Correct Code

```
!pip install yfinance arch -q
import yfinance as yf, numpy as np
import warnings; warnings.filterwarnings('ignore')
from arch import arch_model

btc     = yf.download("BTC-USD", start="2019-01-01",
   end="2024-12-31")
returns = np.log(btc["Close"]).diff().dropna() * 100

fit = arch_model(returns, mean="Constant",
                 vol="GARCH", p=1, q=1,
                    dist="t").fit(disp="off")

omega      = fit.params["omega"]        # intercept of
   variance equation
alpha      = fit.params["alpha[1]"]
beta       = fit.params["beta[1]"]      # "beta[1]" not
   "beta"
nu         = fit.params["nu"]
alpha_beta = alpha + beta
lrv        = omega / (1 - alpha_beta)   # divide by (1 -
   alpha - beta)
lrvol      = np.sqrt(lrv)
```

```python
hl          = np.log(0.5) / np.log(alpha_beta)  # half-life formula

latest_vol   = fit.conditional_volatility.iloc[-1]  # .iloc[-1] for last value
latest_vol_a = latest_vol * np.sqrt(252)

print("="*50)
print("  BTC-USD GARCH(1,1) -- Risk Report Card")
print("="*50)
print(f"  Persistence (a+b)     : {alpha_beta:.4f}")
print(f"  Shock half-life       : {hl:.1f} trading days")
print(f"  Long-run annual vol   : {lrvol*np.sqrt(252):.2f}%")  # 252 trading days
print(f"  Current annual vol    : {latest_vol_a:.2f}%")
print(f"  t degrees of freedom  : {nu:.2f}")
print(f"  Tail classification   : {'Extreme fat tails' if nu<5 else 'Heavy tails'}")
print("="*50)
```

② Plain English

This report card translates raw GARCH parameter estimates into business-readable numbers. Persistence tells risk managers whether a volatility spike will linger (high $\alpha + \beta$) or dissipate quickly (low $\alpha + \beta$). The half-life converts this into intuitive calendar days. The current vs long-run comparison shows whether BTC is currently in a calm or stormy regime.

③ Common Mistake

Using `np.sqrt(365)` instead of `np.sqrt(252)` to annualise daily volatility. Financial conventions use 252 trading days per year, not 365 calendar days. Using 365 overstates annualised volatility by a factor of $\sqrt{365/252} \approx 1.20$ — a 20% error that would make position sizing calculations meaningfully wrong.

④ Real World Link

This exact table appears in crypto exchange risk management dashboards. When Binance or Coinbase quotes "implied volatility" for BTC options, it's derived from a model with these exact inputs. The "current annual vol" number (typically 60–120% for BTC) directly sets options premiums.

M5.005 *Guided Mini-Project* **EASY**

Topic: MRF — Investment Signal from Moving Averages

The Brief: Compute a simple Golden Cross / Death Cross signal for MRF: go long when 50-day MA crosses above 200-day MA, go short (or exit) when it crosses below.

```python
mrf   = yf.download("MRF.NS", start="2019-01-01", end="2024-12-31")
close = mrf["Close"].dropna()
```

```
3
4  ma50  = close.rolling(______).mean()
5  ma200 = close.rolling(______).mean()
6
7  signal = pd.Series(np.where(ma50 > ma200, 1, -1),
8                     index=close.index, name="Signal")
9
10 daily_ret    = np.log(close).______.dropna()
11 strat_ret    = signal.shift(1).reindex(daily_ret.index) *
       daily_ret
12 bh_ret       = daily_ret
13
14 cum_strat    = (1 + strat_ret).______()
15 cum_bh       = (1 + bh_ret  ).______()
16
17 fig, axes = plt.subplots(2,1, figsize=(16,10))
18 axes[0].plot(close,  color="gray",   linewidth=0.8,
       alpha=0.5, label="MRF Close")
19 axes[0].plot(ma50,   color="blue",   linewidth=1.5,
       label="50-Day MA")
20 axes[0].plot(ma200,  color="red",    linewidth=2,
       label="200-Day MA")
21 axes[0].set_title("MRF -- Golden/Death Cross Signal");
       axes[0].legend()
22
23 axes[1].plot(cum_strat, color="green",  linewidth=2,
       label="MA Crossover Strategy")
24 axes[1].plot(cum_bh,    color="orange", linewidth=2,
       label="Buy and Hold")
25 axes[1].set_title("Cumulative Return Comparison");
       axes[1].legend()
26 plt.tight_layout(); plt.show()
27
28 total_strat = cum_strat.______ - 1
29 total_bh    = cum_bh.______ - 1
30 print(f"Strategy total return : {total_strat*100:.1f}%")
31 print(f"Buy-hold total return : {total_bh*100:.1f}%")
32 print(f"Strategy beats B&H    : {total_strat > ______}")
```

Solution

① Correct Code

```
1 !pip install yfinance -q
2 import yfinance as yf, numpy as np, pandas as pd
3 import matplotlib.pyplot as plt, warnings
4 warnings.filterwarnings('ignore')
5
```

```
mrf   = yf.download("MRF.NS", start="2019-01-01",
    end="2024-12-31")
close = mrf["Close"].dropna()

ma50  = close.rolling(50).mean()    # 50-day window
ma200 = close.rolling(200).mean()   # 200-day window

signal = pd.Series(np.where(ma50 > ma200, 1, -1),
                   index=close.index, name="Signal")

daily_ret = np.log(close).diff().dropna()  # .diff() for
    log returns
strat_ret = signal.shift(1).reindex(daily_ret.index) *
    daily_ret  # lag avoids lookahead
bh_ret    = daily_ret

cum_strat = (1 + strat_ret).cumprod()  # .cumprod() for
    cumulative product
cum_bh    = (1 + bh_ret   ).cumprod()

fig, axes = plt.subplots(2,1, figsize=(16,10))
axes[0].plot(close, color="gray", lw=0.8, alpha=0.5,
    label="MRF Close")
axes[0].plot(ma50,  color="blue", lw=1.5, label="50-Day
    MA")
axes[0].plot(ma200, color="red",  lw=2,   label="200-Day
    MA")
axes[0].set_title("MRF -- Golden/Death Cross Signal");
    axes[0].legend()
axes[1].plot(cum_strat, color="green",  lw=2, label="MA
    Crossover Strategy")
axes[1].plot(cum_bh,    color="orange", lw=2, label="Buy
    and Hold")
axes[1].set_title("Cumulative Return Comparison");
    axes[1].legend()
plt.tight_layout(); plt.show()

total_strat = cum_strat.iloc[-1] - 1   # .iloc[-1] for
    final cumulative return
total_bh    = cum_bh.iloc[-1]    - 1
print(f"Strategy total return : {total_strat*100:.1f}%")
print(f"Buy-hold total return : {total_bh*100:.1f}%")
print(f"Strategy beats B&H    : {total_strat > total_bh}")
```

② Plain English

The Golden Cross (50-day crosses above 200-day) has been used by technical analysts for decades. `.shift(1)` is critical — it ensures the signal is based on yesterday's MA relationship, not today's, preventing look-ahead bias. Whether the strategy beats buy-and-hold depends on whether MRF's trend periods are long enough to

generate net profit after whipsaw losses during choppy markets.

③ Common Mistake

Forgetting `.shift(1)` on the signal before multiplying by returns. Without the shift, the strategy looks at today's MA crossover to determine today's return — look-ahead bias (impossible in practice). Always lag signals by one period before applying to returns.

④ Real World Link

Institutional desks use this crossover framework systematically across hundreds of Indian stocks. MRF's behaviour during 2019–2024 (strong uptrend interrupted by COVID crash) means the strategy likely missed some of the uptrend by switching to short/flat too early during the 2020 crash. The cumulative return comparison quantifies this exactly.

Section B — Medium Problems — M5.006 to M5.015

M5.006 *Write from Scratch* MEDIUM

Topic: BTC — Complete ARIMA+GARCH Two-Stage Model

The Brief: A quant at a trading firm says: "Build the standard two-stage model for BTC: ARIMA for the mean, GARCH for the residuals." Write complete code that: (1) fits ARIMA(1,0,1) to BTC log returns, (2) extracts ARIMA residuals, (3) fits GARCH(1,1) to those residuals, (4) generates 10-day return and volatility forecasts, (5) computes 95% forecast intervals combining both, (6) plots the result.

Solution

① Correct Code

```
!pip install yfinance statsmodels arch -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, scipy.stats as stats
import warnings; warnings.filterwarnings('ignore')
from statsmodels.tsa.arima.model import ARIMA
from arch import arch_model

btc     = yf.download("BTC-USD", start="2019-01-01",
   end="2024-12-31")
returns = np.log(btc["Close"]).diff().dropna() * 100

# Stage 1: ARIMA Mean Model
arima_fit    = ARIMA(returns, order=(1,0,1)).fit()
fc_mean      =
   arima_fit.get_forecast(steps=10).predicted_mean.values
arima_resid  = arima_fit.resid.dropna()
```

```
print(f"ARIMA(1,0,1): AIC={arima_fit.aic:.2f}")

# Stage 2: GARCH on ARIMA Residuals (mean="Zero" -- ARIMA already handled mean)
garch_fit  = arch_model(arima_resid, mean="Zero",
                        vol="GARCH", p=1,
                           q=1).fit(disp="off")
garch_fc   = garch_fit.forecast(horizon=10, reindex=False)
fc_vol     = np.sqrt(garch_fc.variance.iloc[0].values)

alpha = garch_fit.params["alpha[1]"]; beta = garch_fit.params["beta[1]"]
print(f"GARCH(1,1): alpha+beta={alpha+beta:.4f}")

# Combined 95% CI
z95      = stats.norm.ppf(0.975)    # 1.96
fc_lower = fc_mean - z95 * fc_vol
fc_upper = fc_mean + z95 * fc_vol

print("\n10-Day Forecast (ARIMA mean + GARCH volatility):")
print(f"  {'Day':>4}  {'Mean(%)':>10}  {'Vol(%)':>10}  {'95% Lower':>12}  {'95% Upper':>12}")
for h in range(10):
    print(f"  {h+1:>4}  {fc_mean[h]:>+10.4f}  {fc_vol[h]:>10.4f}"
          f"  {fc_lower[h]:>12.4f}  {fc_upper[h]:>12.4f}")

# Plot
fig, axes = plt.subplots(2,1, figsize=(16,10))
fig.suptitle("BTC-USD -- ARIMA(1,0,1) + GARCH(1,1) Two-Stage Model", fontsize=13)

hist_plot = returns[-60:].values
axes[0].plot(range(60), hist_plot, color="steelblue", lw=0.8, label="Historical")
axes[0].plot(range(60,70), fc_mean, "r-o", lw=2, markersize=5, label="ARIMA Forecast")
axes[0].fill_between(range(60,70), fc_lower, fc_upper,
                     color="red", alpha=0.2, label="95% CI (GARCH vol)")
axes[0].axhline(0, color="black", lw=0.8, linestyle="--")
axes[0].axvline(60, color="gray", lw=1.5, linestyle=":")
axes[0].set_title("Return Forecast with GARCH-Based Confidence Intervals")
axes[0].legend()

vol_hist = garch_fit.conditional_volatility[-60:].values
axes[1].plot(range(60), vol_hist, color="orange", lw=1.5, label="Historical Conditional Vol")
```

```
axes[1].plot(range(60,70), fc_vol, "go-", lw=2,
    markersize=5, label="GARCH Forecast")
axes[1].axvline(60, color="gray", lw=1.5, linestyle=":")
axes[1].set_title("Conditional Volatility Forecast");
    axes[1].legend()
plt.tight_layout(); plt.show()
```

② Plain English

The two-stage model separates the "direction" question (ARIMA mean model) from the "risk" question (GARCH variance model). ARIMA says where returns are expected to go; GARCH says how uncertain that expectation is. The confidence interval widens as horizon extends — both because ARIMA mean forecast uncertainty grows and because GARCH variance forecast approaches its long-run level.

③ Common Mistake

Fitting GARCH to raw returns instead of ARIMA residuals in Stage 2. If you fit GARCH to raw returns using `mean="Constant"`, the GARCH optimiser estimates its own mean — which may conflict with the ARIMA mean estimate. Using `mean="Zero"` on ARIMA residuals ensures the two stages don't contradict each other.

④ Real World Link

This two-stage structure (ARIMA + GARCH) is used in quantitative trading for BTC position sizing: the ARIMA forecast determines the direction and magnitude of the bet; the GARCH volatility forecast determines how large the position should be (target a fixed percentage of portfolio at risk). This is the foundation of volatility-targeted trading strategies.

M5.007 *Write from Scratch* MEDIUM

Topic: MRF — Holt-Winters vs SARIMA Forecast Horse Race

The Brief: The head of MRF equity research asks: "Which model gives better 12-month price forecasts — Holt-Winters or SARIMA? Show me the numbers." Write complete code that: (1) downloads MRF monthly data, (2) splits into 48-month training and 12-month test, (3) fits both Holt-Winters (best variant by AIC) and SARIMA(1,1,1)(1,1,0)[12], (4) generates 12-month forecasts from each, (5) computes RMSE, MAE, MAPE for both, (6) plots both forecasts against actuals, (7) declares a winner and explains why.

Solution

① Correct Code

```
!pip install yfinance statsmodels -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
```

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing
from statsmodels.tsa.statespace.sarimax import SARIMAX

mrf     = yf.download("MRF.NS", start="2016-01-01", end="2024-12-31")
monthly = mrf["Close"].dropna().resample("ME").last()
train   = monthly.iloc[:-12]; test = monthly.iloc[-12:]
print(f"Train: {len(train)}m | Test: {len(test)}m")

# Holt-Winters -- best variant by AIC
hw_candidates = [
    ("HW-AddAdd", {"trend":"add","seasonal":"add","seasonal_periods":12}),
    ("HW-AddMul", {"trend":"add","seasonal":"mul","seasonal_periods":12}),
    ("Holt",      {"trend":"add","seasonal":None}),
]
best_hw_name, best_hw_aic, best_hw_fit = None, np.inf, None
for name, params in hw_candidates:
    try:
        m = ExponentialSmoothing(train, **params) if params["seasonal"] \
            else ExponentialSmoothing(train, trend=params["trend"])
        f = m.fit(optimized=True, disp=False)
        if f.aic < best_hw_aic:
            best_hw_aic = f.aic; best_hw_name = name; best_hw_fit = f
    except Exception: pass
hw_fc = best_hw_fit.forecast(12)
print(f"Best HW: {best_hw_name} (AIC={best_hw_aic:.2f})")

# SARIMA
sarima_fit = SARIMAX(train, order=(1,1,1), seasonal_order=(1,1,0,12),
                     enforce_stationarity=False, enforce_invertibility=False
                     ).fit(disp=False)
sarima_fc  = sarima_fit.forecast(12)
print(f"SARIMA AIC: {sarima_fit.aic:.2f}")

# Metrics
def metrics(actual, forecast, name):
    e    = actual.values - forecast.values
    rmse = np.sqrt(np.mean(e**2))
    mae  = np.mean(np.abs(e))
    mape = np.mean(np.abs(e/actual.values)) * 100
```

```
    print(f"  {name:28} | RMSE=Rs.{rmse:,.0f} |
       MAPE={mape:.2f}%")
    return rmse, mae, mape

print("\n12-Month Test Performance:")
r_hw, m_hw, p_hw = metrics(test, hw_fc,    best_hw_name)
r_sa, m_sa, p_sa = metrics(test, sarima_fc,
   "SARIMA(1,1,1)(1,1,0)[12]")
winner = best_hw_name if p_hw < p_sa else "SARIMA"
print(f"\n  Winner by MAPE: {winner}")

# Plot
fig, ax = plt.subplots(figsize=(16,6))
monthly[-36:].plot(ax=ax, color="steelblue", lw=1.5,
   label="Historical")
test.plot(ax=ax, color="black", lw=2, linestyle="--",
   label="Actual Test")
hw_fc.plot(ax=ax, color="green", lw=2,
   label=f"{best_hw_name} (MAPE={p_hw:.1f}%)")
sarima_fc.plot(ax=ax, color="red", lw=2, label=f"SARIMA
   (MAPE={p_sa:.1f}%)")
ax.axvline(train.index[-1], color="gray", lw=1.5,
   linestyle=":")
ax.set_title("MRF Monthly -- Holt-Winters vs SARIMA:
   12-Month Forecast Horse Race")
ax.set_ylabel("Rs. INR"); ax.legend(); plt.tight_layout();
   plt.show()
```

② Plain English

A horse race is the only honest way to compare forecasting models — AIC compares in-sample fit, but MAPE on a held-out test set compares real predictive power. SARIMA often wins for series with strong, stable seasonality (captures seasonal structure explicitly). Holt-Winters wins when the seasonal pattern is irregular or when n is too small for SARIMA to estimate seasonal parameters reliably.

③ Common Mistake

Comparing models using their in-sample AIC values — these aren't comparable across different model classes (SARIMA AIC $\neq$ Holt-Winters AIC because they use different likelihood functions). Always compare out-of-sample on a held-out test set using scale-free metrics like MAPE.

④ Real World Link

MRF equity research teams would use exactly this framework before publishing a 12-month price target. If SARIMA wins by 3 MAPE points, that's meaningful — a 3% more accurate forecast on a Rs. 120,000 stock means Rs. 3,600 less forecast error per share.

M5.008 *Write from Scratch* **MEDIUM**

Topic: Gold — Full Stationarity Investigation with Structural Break
The Brief: Gold's price broke structurally during COVID (March 2020). Investigate this statistically.
Write code that: (1) runs ADF and KPSS on Gold log returns before and after March 2020, (2) runs a Chow test for structural break at that date, (3) fits separate AR(1) models to pre- and post-break periods, (4) compares AR(1) coefficients, (5) recommends whether to use full-period or post-2020 data for forecasting.

Solution

① Correct Code

```
!pip install yfinance statsmodels -q
import yfinance as yf, numpy as np, pandas as pd
import scipy.stats as spstats, warnings
warnings.filterwarnings('ignore')
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.tsa.ar_model import AutoReg

gold    = yf.download("GC=F", start="2019-01-01",
   end="2024-12-31")
returns = np.log(gold["Close"]).diff().dropna() * 100
break_date = "2020-04-01"
pre  = returns[:break_date]; post = returns[break_date:]

print("="*60)
print("  Gold Returns -- Structural Break Analysis at
   COVID")
print("="*60)

# ADF + KPSS for both periods
for period, name in [(pre,"Pre-COVID"),(post,"Post-COVID
   (2020-2024)")]:
    adf    = adfuller(period)
    kpss_r = kpss(period, regression="c", nlags="auto")
    kpss_rej = kpss_r[0] > kpss_r[3]["5%"]
    print(f"\n  {name} (n={len(period)}):")
    print(f"    ADF  p : {adf[1]:.6f} -> {'Stationary' if
       adf[1]<0.05 else 'Non-stationary'}")
    print(f"    KPSS   : {'Non-stationary' if kpss_rej
       else 'Stationary'}")

# Chow Test
def ar1_sse(series):
    fit = AutoReg(series, lags=1).fit()
    return np.sum(fit.resid**2), len(series)

sse_pre,  n_pre  = ar1_sse(pre)
```

```
sse_post, n_post = ar1_sse(post)
full_fit = AutoReg(returns, lags=1).fit()
sse_full = np.sum(full_fit.resid**2); n_full =
    len(returns); k = 2

chow_f = ((sse_full-sse_pre-sse_post)/k) /
    ((sse_pre+sse_post)/(n_full-2*k))
p_chow = 1 - spstats.f.cdf(chow_f, dfn=k, dfd=n_full-2*k)

print(f"\n  Chow Test (break at {break_date}):")
print(f"    F-stat  : {chow_f:.4f}")
print(f"    p-value : {p_chow:.4f}")
print(f"    Break detected: {p_chow < 0.05}")

# AR(1) coefficients
fit_pre  = AutoReg(pre,  lags=1).fit()
fit_post = AutoReg(post, lags=1).fit()
ar_pre   = fit_pre.params.iloc[1]; ar_post =
    fit_post.params.iloc[1]
print(f"\n  AR(1) Coefficients:")
print(f"    Pre-COVID  : {ar_pre:.6f}")
print(f"    Post-COVID : {ar_post:.6f}  (change:
    {ar_post-ar_pre:+.6f})")

print(f"\n  Recommendation:")
if p_chow < 0.05:
    print("  Structural break CONFIRMED -- use POST-2020
        data for forecasting.")
    print("  Pre-COVID Gold dynamics differ structurally
        from current regime.")
else:
    print("  No significant structural break -- full
        period data appropriate.")
```

② Plain English

The Chow test is the statistical confirmation of what a chart might suggest visually. If the AR coefficient for Gold returns changed significantly after COVID, that's a regime change. Separate models for each regime are more accurate than one model averaging across structurally different periods.

③ Common Mistake

Choosing the break date by looking at the data first (the biggest spike) and testing at exactly that point. This "data snooping" inflates the false positive rate. Proper practice: specify break dates based on economic events before looking at test results.

④ Real World Link

Gold's structural dynamics genuinely changed post-COVID. The correlation between Gold and real interest rates (historically negative and strong) weakened as central banks floored rates. A model trained on 2019 data would have predicted Gold to

fall when rates rose in 2022 — it rose instead. This is the regime change the Chow test detects.

M5.009 *Write from Scratch* MEDIUM

Topic: BTC — Volatility Regime-Switching Position Sizing
The Brief: Build a simple volatility-targeting position sizing system for BTC: hold a fixed amount of portfolio-level risk (target 20% annualised portfolio vol) by adjusting position size inversely with GARCH conditional volatility.
Write complete code that: (1) fits GARCH to BTC returns, (2) computes daily position size as `target_vol / (current_vol × sqrt(252))`, (3) computes strategy returns, (4) compares Sharpe ratio of vol-targeted vs unscaled BTC, (5) plots position sizes over time alongside BTC price.

Solution

① Correct Code

```
!pip install yfinance arch -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from arch import arch_model

btc      = yf.download("BTC-USD", start="2019-01-01",
   end="2024-12-31")
close    = btc["Close"].dropna()
returns = np.log(close).diff().dropna() * 100

fit = arch_model(returns, mean="Constant", vol="GARCH",
   p=1, q=1).fit(disp="off")
cond_vol_annual = fit.conditional_volatility *
   np.sqrt(252)  # annualised %

target_vol_pct  = 20.0  # target 20% annual portfolio
   volatility
position        = (target_vol_pct /
   cond_vol_annual).clip(0, 2)  # cap at 2x leverage
position_lagged =
   position.shift(1).reindex(returns.index).dropna()
returns_aligned = returns.reindex(position_lagged.index)

strat_returns = position_lagged * returns_aligned / 100
bh_returns    = returns_aligned / 100

def sharpe(ret, ann=252):
    return (ret.mean() / ret.std()) * np.sqrt(ann)
def max_dd(cum):
```

```
    return ((cum - cum.cummax()) / cum.cummax()).min()

cum_strat = (1 + strat_returns).cumprod()
cum_bh    = (1 + bh_returns).cumprod()
sh_strat  = sharpe(strat_returns); sh_bh = sharpe(bh_returns)

print("="*55)
print("  BTC -- Vol-Targeting vs Buy-and-Hold")
print("="*55)
print(f"  {'Metric':25} {'Vol-Target':>12} {'Buy-Hold':>12}")
print(f"  {'-'*50}")
print(f"  {'Sharpe':25} {sh_strat:>12.4f} {sh_bh:>12.4f}")
print(f"  {'Total Return (%)':25} {(cum_strat.iloc[-1]-1)*100:>11.1f}%"
      f" {(cum_bh.iloc[-1]-1)*100:>11.1f}%")
print(f"  {'Max Drawdown (%)':25} {max_dd(cum_strat)*100:>11.1f}%"
      f" {max_dd(cum_bh)*100:>11.1f}%")
print("="*55)

fig, axes = plt.subplots(3,1, figsize=(16,14))
axes[0].plot(close.reindex(position_lagged.index), color="orange", lw=1)
axes[0].set_title("BTC Price"); axes[0].set_ylabel("USD")
ax2 = axes[0].twinx()
ax2.plot(position_lagged, color="blue", lw=0.8, alpha=0.7, label="Position Size")
ax2.axhline(1, color="gray", lw=0.8, linestyle="--")
ax2.set_ylabel("Position (1.0=unlevered)"); ax2.legend(loc="upper right")
axes[1].plot(cond_vol_annual.reindex(position_lagged.index),
             color="red", lw=1, label="GARCH Annual Vol (%)")
axes[1].axhline(target_vol_pct, color="green", lw=1.5, linestyle="--",
                label=f"Target: {target_vol_pct}%")
axes[1].set_title("GARCH Conditional Volatility vs Target"); axes[1].legend()
axes[2].plot(cum_strat, color="green", lw=2, label=f"Vol-Target (Sharpe={sh_strat:.2f})")
axes[2].plot(cum_bh,    color="orange",lw=2, label=f"Buy-Hold (Sharpe={sh_bh:.2f})")
axes[2].set_title("Cumulative Performance"); axes[2].legend()
plt.tight_layout(); plt.show()
```

② Plain English

Vol-targeting is a systematic risk management rule: when GARCH says BTC is in a 100% annual vol regime (a 2× typical level), halve the position to maintain the same portfolio risk. The strategy's Sharpe ratio improvement over buy-and-hold comes from this risk-adjusted sizing — it doesn't necessarily produce higher absolute returns, but more return per unit of risk.

③ Common Mistake

Forgetting to lag the position by one day. Using today's GARCH vol to size today's position requires knowing today's return to compute today's vol — which isn't available until after trading. Always use yesterday's GARCH vol estimate (`shift(1)`) to determine today's position.

④ Real World Link

All major systematic hedge funds (Winton, Man AHL, Two Sigma) use volatility-targeting as a core risk management tool. For crypto specifically, vol-targeting during the 2022 BTC bear market (vol spiked to 150%+ annual) would have automatically reduced BTC exposure to about 13% of the target-vol-driven position — dramatically limiting drawdowns.

M5.010 *Write from Scratch* MEDIUM

Topic: Multi-Asset Portfolio — Rolling Correlation Heatmap

The Brief: A portfolio manager wants a "correlation dashboard" showing how BTC, MRF, and Gold have been moving together or apart over 2019–2024, using rolling 90-day correlations.

Write complete code for a full dashboard: (1) rolling 90-day pairwise correlations time series, (2) snapshot heatmaps at 4 key dates (COVID crash, 2021 bull, FTX collapse, 2024 halving), (3) static full-period heatmap, (4) summary statistics table.

Solution

① Correct Code

```
!pip install yfinance -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')

assets = {"BTC":"BTC-USD", "MRF":"MRF.NS", "Gold":"GC=F"}
ret_dict = {}
for name, ticker in assets.items():
    raw = yf.download(ticker, start="2019-01-01",
        end="2024-12-31", progress=False)
    ret_dict[name] = np.log(raw["Close"]).diff().dropna()
        * 100

```

```
returns = pd.DataFrame(ret_dict).dropna()
window  = 90
pairs   = [("BTC","MRF"),("BTC","Gold"),("MRF","Gold")]
roll_corrs = {f"{a}-{b}": returns[a].rolling(window).corr(returns[b]) for a,b in pairs}
roll_df    = pd.DataFrame(roll_corrs).dropna()
static_corr = returns.corr()

snap_dates  = ["2020-03-16","2021-04-01","2022-11-10","2024-06-01"]
snap_labels = ["COVID Crash\n(Mar 2020)","Crypto Bull\n(Apr 2021)",
               "FTX Collapse\n(Nov 2022)","Halving Rally\n(Jun 2024)"]

fig = plt.figure(figsize=(20, 16))
gs  = fig.add_gridspec(3, 4, hspace=0.4, wspace=0.3)

ax_top = fig.add_subplot(gs[0, :])
for pair, cs in roll_corrs.items():
    ax_top.plot(cs, linewidth=1.5, label=pair)
ax_top.axhline(0, color="black", lw=0.8)
ax_top.set_title("Rolling 90-Day Pairwise Correlations", fontsize=12)
ax_top.set_ylabel("Correlation rho"); ax_top.legend(); ax_top.set_ylim(-1,1)
for snap, label in zip(snap_dates, snap_labels):
    try:
        snap_ts = pd.Timestamp(snap)
        idx = roll_df.index.get_indexer([snap_ts], method="nearest")[0]
        ax_top.axvline(roll_df.index[idx], color="gray", lw=0.8, linestyle=":")
        ax_top.annotate(label, xy=(roll_df.index[idx], 0.9), fontsize=7,
                        ha="center", bbox=dict(boxstyle="round,pad=0.2",
                        facecolor="lightyellow"))
    except Exception: pass

cmap = plt.cm.RdYlGn
for col_idx, (snap, label) in enumerate(zip(snap_dates, snap_labels)):
    ax = fig.add_subplot(gs[1, col_idx])
    snap_ts = pd.Timestamp(snap)
    nearest = roll_df.index[roll_df.index.get_indexer([snap_ts], method="nearest")[0]]
```

```
    sr = roll_df.loc[nearest]
    snap_mat = np.array([[1.0, sr["BTC-MRF"], sr["BTC-Gold"]],
                         [sr["BTC-MRF"], 1.0, sr["MRF-Gold"]],
                         [sr["BTC-Gold"],sr["MRF-Gold"], 1.0]])
    ax.imshow(snap_mat, cmap=cmap, vmin=-1, vmax=1)
    for name in ["BTC","MRF","Gold"]:
        ax.set_xticks([0,1,2]); ax.set_xticklabels(["BTC","MRF","Gold"], fontsize=8)
        ax.set_yticks([0,1,2]); ax.set_yticklabels(["BTC","MRF","Gold"], fontsize=8)
    ax.set_title(label, fontsize=9)
    for i in range(3):
        for j in range(3):
            ax.text(j, i, f"{snap_mat[i,j]:.2f}", ha="center", va="center",
                    fontsize=9, color="white" if abs(snap_mat[i,j])>0.5 else "black")

ax_static = fig.add_subplot(gs[2, :2])
im2 = ax_static.imshow(static_corr.values, cmap=cmap, vmin=-1, vmax=1)
ax_static.set_xticks([0,1,2]); ax_static.set_xticklabels(["BTC","MRF","Gold"])
ax_static.set_yticks([0,1,2]); ax_static.set_yticklabels(["BTC","MRF","Gold"])
ax_static.set_title("Static Full-Period Correlation (2019-2024)")
plt.colorbar(im2, ax=ax_static)
for i in range(3):
    for j in range(3):
        ax_static.text(j, i, f"{static_corr.values[i,j]:.3f}",
                       ha="center", va="center", fontsize=11)

ax_summary = fig.add_subplot(gs[2, 2:])
ax_summary.axis("off")
txt = "Correlation Summary\n" + "-"*28 + "\n"
for pair, cr in roll_corrs.items():
    txt += f"{pair}: mean={cr.mean():.3f}, max={cr.max():.3f}, min={cr.min():.3f}\n"
txt += "\nKey finding: Crisis periods (COVID, FTX)\nshow correlation spikes -- diversification\nfails exactly when needed most."
```

```
ax_summary.text(0.05, 0.95, txt,
    transform=ax_summary.transAxes,
                fontsize=9, va="top",
                    fontfamily="monospace",
                bbox=dict(boxstyle="round",
                    facecolor="lightyellow"))

fig.suptitle("BTC - MRF - Gold -- Dynamic Correlation
    Dashboard", fontsize=14)
plt.show()
print("\nCorrelation Summary:")
for pair, cr in roll_corrs.items():
    print(f"  {pair}: mean={cr.mean():.4f},
        max={cr.max():.4f}, min={cr.min():.4f}")
```

② Plain English

Rolling correlations reveal that static correlations hide crucial dynamics. BTC-Gold might show $\rho = 0.1$ over the full period, but during the COVID crash both spiked to $\rho = 0.6$ (panic selling everything) before returning to near-zero. This "correlation breakdown in crises" is the most dangerous portfolio risk — diversification fails exactly when you need it most.

③ Common Mistake

Using a 30-day window for rolling correlations on daily data. With $n = 30$ and 3 variables, each correlation estimate has enormous sampling variance. Use 60–120 day windows for stable daily correlation estimates. The 90-day window here balances responsiveness with stability.

④ Real World Link

This dashboard is used by every multi-asset portfolio manager running crypto + traditional assets. The key insight: BTC-MRF correlation (typically near zero) determines whether adding BTC diversifies an MRF equity portfolio. Dynamic correlation management is the professional evolution beyond static Markowitz portfolio theory.

M5.011 *Find the Error* MEDIUM

Topic: End-to-End Pipeline — Three Hidden Bugs

The Brief: A junior analyst handed you this "complete Gold GARCH analysis." It runs without crashing but contains **three conceptual errors** that would make the output misleading. Find all three.

```
from arch import arch_model

gold    = yf.download("GC=F", start="2019-01-01",
    end="2024-12-31")
monthly = gold["Close"].dropna().resample("ME").last()    #
    ERROR 1

```

```
returns = np.log(monthly).diff().dropna()    # ERROR 2: no scaling

model = arch_model(returns, mean="Constant", vol="GARCH", p=1, q=1)
fit   = model.fit(disp="off")

# Annualise volatility
cond_vol_annual = fit.conditional_volatility * 252    # ERROR 3

print(f"Estimated annual vol: {cond_vol_annual.mean():.2f}%")
print(f"Alpha + Beta: {fit.params['alpha[1]'] + fit.params['beta[1]']:.4f}")
```

Solution

① Correct Code

```
!pip install yfinance arch -q
import yfinance as yf, numpy as np
import warnings; warnings.filterwarnings('ignore')
from arch import arch_model

# ERROR 1 FIX: Use DAILY data for GARCH, not monthly
# GARCH is designed for high-frequency data (daily minimum).
# Monthly data has only ~72 observations -- far too few
# for GARCH to reliably estimate volatility dynamics.
gold  = yf.download("GC=F", start="2019-01-01", end="2024-12-31")
close = gold["Close"].dropna()    # DAILY data

# ERROR 2 FIX: Multiply by 100 to get percentage returns
# arch library operates best on percentage-scale data (~1-5%).
# Decimal returns (~0.01) cause numerical precision issues --
# the variance intercept omega becomes ~1e-7 making convergence unstable.
returns = np.log(close).diff().dropna() * 100    # add x 100

model = arch_model(returns, mean="Constant", vol="GARCH", p=1, q=1)
fit   = model.fit(disp="off")

```

```python
# ERROR 3 FIX: Annualise by multiplying by sqrt(252), NOT 252
# Standard deviation (volatility) scales with the SQUARE ROOT of time:
#   Annual vol = Daily vol x sqrt(252)
# Multiplying by 252 gives annual VARIANCE, not annual VOLATILITY.
# For Gold with ~1% daily vol: 252 gives "252%" vs correct ~16%.
cond_vol_annual = fit.conditional_volatility * np.sqrt(252)   # sqrt(252)

print("="*50)
print("  Gold GARCH -- Corrected Analysis")
print("="*50)
print(f"  Daily observations    : {len(returns)}")
print(f"  Mean annual vol       : {cond_vol_annual.mean():.2f}%")
print(f"  Alpha + Beta          : {fit.params['alpha[1]'] + fit.params['beta[1]']:.4f}")
print(f"\n  Three errors fixed:")
print(f"  1. Using daily (not monthly) data")
print(f"  2. Scaling returns by x100 (percentage)")
print(f"  3. Annualising vol by x sqrt(252) (not x 252)")
```

② Plain English

Three independent errors, each producing a wrong result. Error 1 is a data frequency mismatch — GARCH on monthly data would estimate extreme persistence from just 72 data points, completely unreliable. Error 2 causes numerical instability. Error 3 is the most consequential: multiplying by 252 instead of $\sqrt{252}$ would make a 1% daily vol appear as "252%" annualised — off by a factor of 16.

③ Common Mistake

Error 3 is particularly common because students confuse variance scaling ($\times T$) with volatility scaling ($\times\sqrt{T}$). Variance adds linearly over time: 10-day variance = 10 $\times$ 1-day variance. But standard deviation scales with $\sqrt{T}$: 10-day vol = $\sqrt{10}$ $\times$ 1-day vol. This square root of time rule is one of the most fundamental and most frequently violated rules in quantitative finance.

④ Real World Link

Error 3 (multiplying by 252 instead of $\sqrt{252}$) is a career-ending mistake if caught by a senior risk manager. A junior analyst reporting "Gold's current annualised volatility is 252%" when the correct answer is "16%" would immediately raise red flags about their quantitative competency.

M5.012 *Write from Scratch* MEDIUM

Topic: BTC — Auto ARIMA with Walk-Forward Validation
The Brief: Use `pmdarima`'s `auto_arima` to find the best ARIMA order automatically, then validate it with walk-forward testing.
Write complete code that: (1) uses `auto_arima` to select best ARIMA order, (2) reports selected order and AIC, (3) implements 90-day walk-forward validation, (4) computes and plots directional accuracy, (5) tests whether directional accuracy is significantly above 50% using a binomial test.

Solution

① **Correct Code**

```
!pip install yfinance pmdarima statsmodels -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from pmdarima import auto_arima
from scipy.stats import binom_test

btc      = yf.download("BTC-USD", start="2019-01-01",
   end="2024-12-31")
returns = np.log(btc["Close"]).diff().dropna() * 100

print("Running auto_arima...")
auto_fit   = auto_arima(returns, start_p=0, max_p=3,
   start_q=0, max_q=3, d=0,
                          information_criterion="aic",
                             stepwise=True,
                          suppress_warnings=True,
                             error_action="ignore")
best_order = auto_fit.order
p, d, q    = best_order
print(f"Selected: ARIMA{best_order}
   (AIC={auto_fit.aic():.2f})")
print(auto_fit.summary())

# Walk-Forward Directional Accuracy
ret_arr   = returns.values; n = len(ret_arr); min_train =
   252
fc_signs  = []; act_signs = []; fc_vals = []

for t in range(min_train, n - 1):
    try:
        wf = auto_arima(ret_arr[0:t], order=(p,d,q),
                        suppress_warnings=True,
                           error_action="ignore")
        fc = wf.predict(n_periods=1)[0]
    except Exception:
```

```
        fc = 0.0
    fc_vals.append(fc)
    fc_signs.append(1 if fc > 0 else -1)
    act_signs.append(1 if ret_arr[t] > 0 else -1)

fc_signs  = np.array(fc_signs); act_signs = np.array(act_signs)
dir_acc   = np.mean(fc_signs == act_signs)
correct   = (fc_signs == act_signs).astype(float)
roll_acc  = pd.Series(correct).rolling(90).mean()

n_correct = int(dir_acc * len(act_signs))
p_binom   = binom_test(n_correct, len(act_signs), 0.5, alternative="greater")

print(f"\nWalk-Forward Directional Accuracy:")
print(f"  Steps           : {len(act_signs)}")
print(f"  Correct         : {n_correct}/{len(act_signs)}")
print(f"  Accuracy        : {dir_acc*100:.2f}%  (vs coin flip 50%)")
print(f"  Binomial p      : {p_binom:.4f}")
print(f"  Significantly >50%? {p_binom < 0.05}")

fig, axes = plt.subplots(2,1, figsize=(16,8))
axes[0].plot(roll_acc.values*100, color="purple", lw=1.2)
axes[0].axhline(50, color="red", lw=1.5, linestyle="--", label="Coin flip (50%)")
axes[0].axhline(dir_acc*100, color="green", lw=1.5, linestyle="--",
                label=f"Overall: {dir_acc*100:.1f}%")
axes[0].set_title(f"Rolling 90-Day Directional Accuracy -- ARIMA{best_order}")
axes[0].set_ylabel("Accuracy (%)"); axes[0].legend(); axes[0].set_ylim(30,70)

axes[1].scatter(fc_vals,
    ret_arr[min_train:min_train+len(fc_vals)],
                alpha=0.2, s=5, color="steelblue")
axes[1].axhline(0, color="gray", lw=0.8); axes[1].axvline(0, color="gray", lw=0.8)
axes[1].set_title(f"ARIMA{best_order} -- Forecast vs Actual Returns")
axes[1].set_xlabel("Forecasted Return (%)"); axes[1].set_ylabel("Actual Return (%)")
plt.tight_layout(); plt.show()
```

② Plain English

Directional accuracy measures what matters for trading — not whether the model got the exact return right, but whether it got the sign right. A coin flip gives 50%.

For BTC, we expect directional accuracy close to 50% — confirming that the mean process is close to random walk and GARCH (not ARIMA) is the value-add for BTC modelling.

③ Common Mistake

Not testing whether directional accuracy is statistically significantly above 50% — just reporting the raw percentage. Getting 51% correct in 1000 trials is not meaningful (could easily be chance). The binomial test properly accounts for sampling variance. With $n = 700$ walk-forward steps, you need at least 53–54% to be statistically significant at 5%.

④ Real World Link

"Directional accuracy" is the metric that algorithmic trading desks care about most — it translates directly into win rate on a directional strategy. The empirical finding (BTC ARIMA directional accuracy $\approx 50\%$) is precisely why crypto quant funds focus on volatility strategies (GARCH) rather than directional strategies (ARIMA).

M5.013 *Write from Scratch* MEDIUM

Topic: Gold — Complete Residual Diagnostics Suite as a Function

The Brief: Write a production-quality diagnostic function that works for ANY fitted statsmodels/arch model and produces a comprehensive residual report.

Write a function `model_diagnostics(resid, model_name, is_garch)` that produces: (1) 6-panel diagnostic figure (residual series, ACF, PACF, histogram vs Normal, Q-Q plot, ACF of squared residuals), (2) five statistical tests (Ljung-Box, ARCH LM, Jarque-Bera, Shapiro-Wilk, skewness/kurtosis), (3) plain-English verdict and recommendations.

Solution

① Correct Code

```
!pip install yfinance statsmodels arch -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, scipy.stats as spstats
import warnings; warnings.filterwarnings('ignore')
from statsmodels.graphics.tsaplots import plot_acf,
    plot_pacf
from statsmodels.stats.diagnostic import acorr_ljungbox
from arch.univariate import arch_lm_test
from arch import arch_model
from statsmodels.tsa.arima.model import ARIMA

def model_diagnostics(resid, model_name="Model",
    is_garch=False):
    """Complete residual diagnostic suite. Returns report
        dict."""
    resid = pd.Series(resid).dropna().values
```

```
14
15      # Tests
16      lb    = acorr_ljungbox(resid, lags=20, return_df=True)
17      lb_min= lb["lb_pvalue"].min()
18      lm    = arch_lm_test(pd.Series(resid), nlags=10); lm_p
           = lm.pvalue
19      jb_p  = spstats.jarque_bera(resid).pvalue
20      sw_p  = spstats.shapiro(resid[:500])[1]
21      skew  = spstats.skew(resid); kurt =
           spstats.kurtosis(resid)
22
23      # 6-Panel Figure
24      fig, axes = plt.subplots(2, 3, figsize=(18, 10))
25      fig.suptitle(f"Residual Diagnostics -- {model_name}",
           fontsize=13, fontweight="bold")
26
27      axes[0,0].plot(resid, color="steelblue", lw=0.6)
28      axes[0,0].axhline(0, color="red", lw=1, linestyle="--")
29      axes[0,0].axhline( 2*resid.std(), color="orange",
           lw=1, linestyle=":")
30      axes[0,0].axhline(-2*resid.std(), color="orange",
           lw=1, linestyle=":")
31      axes[0,0].set_title("Residuals Over Time (orange =
           +/-2-sigma)")
32
33      plot_acf(resid, lags=30, ax=axes[0,1], zero=False,
           alpha=0.05)
34      axes[0,1].set_title(f"ACF of Residuals\nLB min
           p={lb_min:.4f} "
35                          f"({'OK' if lb_min>0.05 else
                               'Autocorrelated!'})")
36
37      plot_pacf(resid, lags=30, ax=axes[0,2], zero=False,
           alpha=0.05)
38      axes[0,2].set_title("PACF of Residuals")
39
40      axes[1,0].hist(resid, bins=60, density=True,
           color="steelblue", alpha=0.6)
41      x = np.linspace(resid.min(), resid.max(), 300)
42      axes[1,0].plot(x, spstats.norm.pdf(x, resid.mean(),
           resid.std()), "r-", lw=2)
43      axes[1,0].set_title(f"Distribution\nSkew={skew:.3f},
           ExKurt={kurt:.3f}")
44
45      spstats.probplot(resid, dist="norm", plot=axes[1,1])
46      axes[1,1].set_title("Q-Q Plot (deviations =
           non-Gaussian)")
47
```

```
    plot_acf(resid**2, lags=30, ax=axes[1,2], zero=False, alpha=0.05, color="orange")
    axes[1,2].set_title(f"ACF Squared Residuals\nARCH LM p={lm_p:.4f} "
                        f"({'Clean' if lm_p>0.05 else 'ARCH effects!'})")

    plt.tight_layout(); plt.show()

    report = {"lb_min_pvalue":lb_min,"arch_lm_pvalue":lm_p,
              "jb_pvalue":jb_p,"sw_pvalue":sw_p,
              "skewness":skew,"excess_kurtosis":kurt}
    print(f"\n{'='*55}")
    print(f"  Residual Report -- {model_name}")
    print(f"{'='*55}")
    print(f"  Ljung-Box min p (mean autocorr) : {lb_min:.4f} "
          f"{'OK' if lb_min>0.05 else '-- Autocorrelation remains -> increase order'}")
    print(f"  ARCH LM p (var autocorr)        : {lm_p:.4f} "
          f"{'OK' if lm_p>0.05 else '-- ARCH effects -> add GARCH'}")
    print(f"  Skewness / Excess kurtosis      : {skew:.4f} / {kurt:.4f}")
    overall = (lb_min > 0.05) and (lm_p > 0.05)
    print(f"\n  Overall: {'ADEQUATE' if overall else 'NEEDS REVISION'}")
    print(f"{'='*55}\n")
    return report

# Apply to Gold ARIMA and GARCH
gold    = yf.download("GC=F", start="2019-01-01", end="2024-12-31")
returns = np.log(gold["Close"]).diff().dropna() * 100

arima_fit = ARIMA(returns, order=(1,0,1)).fit()
r1 = model_diagnostics(arima_fit.resid.dropna(), "ARIMA(1,0,1) -- Gold Returns")

garch_fit = arch_model(arima_fit.resid.dropna(), mean="Zero",
                       vol="GARCH", p=1, q=1).fit(disp="off")
r2 = model_diagnostics(garch_fit.std_resid.dropna(),
                       "GARCH(1,1) Std Residuals -- Gold")
```

② Plain English

This function encapsulates the complete residual validation workflow. The key logic: if Ljung-Box on residuals is significant → ARIMA order is wrong; if ARCH LM on residuals is significant → GARCH is needed. Both conditions should be satisfied for a well-specified model.

③ Common Mistake

For GARCH models, running diagnostics on raw residuals (ε_t) instead of standardised residuals ($z_t = \varepsilon_t/\sigma_t$). Raw GARCH residuals still show ARCH effects by construction. Always pass `fit.std_resid` (not `fit.resid`) to this function for GARCH models.

④ Real World Link

This function should be the final step of every model submitted to a risk committee. Professional model documentation requires precisely these six diagnostic elements. Having them in a reusable function ensures consistency across all models in a portfolio with zero additional coding effort.

M5.014 *Predict the Output* MEDIUM

Topic: MRF — Understanding GARCH Parameter Interaction

Without running this code, predict the qualitative shape of the conditional volatility and answer the interpretation questions.

```
from arch import arch_model
import numpy as np, pandas as pd

np.random.seed(42)
calm_ret  = np.random.randn(200) * 1.0   # 1% daily vol
storm_ret = np.random.randn(60)  * 5.0   # 5% daily vol (crisis)
calm2_ret = np.random.randn(200) * 1.0   # 1% daily vol again
returns   = pd.Series(np.concatenate([calm_ret, storm_ret, calm2_ret]))

fit = arch_model(returns, mean="Zero", vol="GARCH", p=1, q=1).fit(disp="off")
a1  = fit.params["alpha[1]"]; b1 = fit.params["beta[1]"]

print(f"alpha={a1:.4f}, beta={b1:.4f}, alpha+beta={a1+b1:.4f}")

cv = fit.conditional_volatility
print(f"Vol in calm period   (days 1-200)  : {cv[:200].mean():.4f}")
print(f"Vol in storm period  (days 201-260): {cv[200:260].mean():.4f}")
print(f"Vol in recovery      (days 261-460):
```

```
    {cv[260:].mean():.4f}")
print(f"Storm/calm vol ratio                :
    {cv[200:260].mean()/cv[:200].mean():.2f}x")
print(f"Recovery/calm vol ratio             :
    {cv[260:].mean()/cv[:200].mean():.2f}x")
```

Predict: (i) approximate alpha+beta? (ii) recovery time? (iii) recovery/-calm vol ratio?

Solution

① Predicted Output

```
alpha≈0.10, beta≈0.85, alpha+beta≈0.95
Vol calm (1-200) :  ≈1.2--1.6
Vol storm (201-260):  ≈3.5--5.5 (3--4× calm)
Vol recovery :  ≈1.2--1.8 (near calm)
Storm/calm ratio :  ≈ 3--5×
Recovery/calm ratio:  ≈ 1.0--1.3× (near calm after 200 days)
```

```
# Reasoning:
# Data has genuine volatility clustering (calm -> storm -> calm)
# GARCH will capture this with moderate-to-high persistence.
# alpha+beta ~= 0.95 (typical for financial-like data with clustering)
#
# Half-life = log(0.5)/log(0.95) ~= 13.5 days
# After the storm ends at day 260, 200 days of recovery remains.
# 200 days >> 4 half-lives --> vol should be very close to baseline
# Recovery/calm ratio: close to 1.0 (but slightly above due to the
# storm shifting the estimated long-run variance upward)
#
# Verification (run this to confirm):
import numpy as np, pandas as pd
from arch import arch_model
np.random.seed(42)
r  = pd.Series(np.concatenate([np.random.randn(200)*1.0,
                               np.random.randn(60)*5.0,
                               np.random.randn(200)*1.0]))
f  = arch_model(r, mean="Zero", vol="GARCH", p=1, q=1).fit(disp="off")
a1 = f.params["alpha[1]"]; b1 = f.params["beta[1]"]
hl = np.log(0.5)/np.log(a1+b1)
cv = f.conditional_volatility
```

```
print(f"alpha={a1:.4f}, beta={b1:.4f},
    half-life={hl:.1f}d")
print(f"Storm/calm:
    {cv[200:260].mean()/cv[:200].mean():.2f}x")
print(f"Recovery/calm:
    {cv[260:].mean()/cv[:200].mean():.2f}x")
```

② Plain English

This problem builds intuition for GARCH's key behaviour: mean reversion of volatility. The storm (5× elevated vol) creates a large shock that GARCH captures immediately. After the storm ends, conditional vol decays back toward baseline at the exponential rate determined by $\alpha + \beta$. With $\alpha + \beta \approx 0.95$ and 200-day recovery window, the model should largely have returned to calm levels.

③ Common Mistake

Predicting recovery/calm ratio = 1.0 exactly. GARCH uses the full history of returns, and the storm period (60 days) shifts the estimated long-run variance slightly. The unconditional variance $\omega/(1-\alpha-\beta)$ is estimated from all 460 observations, so it's slightly above the true calm-period variance. Expect recovery ratio slightly above 1.0 (e.g. 1.1–1.3).

④ Real World Link

This simulation shows exactly how BTC's GARCH model would have behaved after the COVID crash and FTX collapse. With persistence $\alpha+\beta \approx 0.96$, the model would have predicted elevated volatility for 2–3 months post-crash — which matches the historical observation that BTC options implied vol remained elevated until June 2020 and February 2023 respectively.

M5.015 *Write from Scratch* MEDIUM

Topic: MRF — Rolling ARIMA with Model Drift Detection
The Brief: Fit ARIMA(p,0,q) to MRF returns using a rolling 252-day window and track whether the selected order and AIC drift over time.
Write complete code that: (1) implements rolling 252-day window ARIMA with order selection from candidates $(p, q) \in \{(0,0), (1,0), (0,1), (1,1)\}$, (2) records date, selected (p, q), AIC, AR coefficient at each window, (3) plots how order selection and AIC evolve over 2019–2024, (4) identifies periods of highest instability.

Solution

① Correct Code

```
!pip install yfinance statsmodels -q
import yfinance as yf, numpy as np, pandas as pd
```

```
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from statsmodels.tsa.arima.model import ARIMA

mrf     = yf.download("MRF.NS", start="2018-01-01", end="2024-12-31")
returns = np.log(mrf["Close"]).diff().dropna() * 100
ret_arr = returns.values; dates = returns.index
n = len(ret_arr); window = 252
candidates = [(0,0),(1,0),(0,1),(1,1)]; results = []

for t in range(window, n):
    train = ret_arr[t-window:t]; best_aic = np.inf; best_order = (0,0); best_ar = np.nan
    for (p, q) in candidates:
        try:
            fit = ARIMA(train, order=(p,0,q)).fit()
            if fit.aic < best_aic:
                best_aic = fit.aic; best_order = (p,q)
                best_ar  = fit.params.get("ar.L1", np.nan) if p == 1 else np.nan
        except Exception: pass
    results.append({"date":dates[t],"p":best_order[0],"q":best_order[1],
                    "pq_label":f"({best_order[0]},{best_order[1]})",
                    "AIC":best_aic,"ar_coef":best_ar})

df = pd.DataFrame(results).set_index("date")
df["order_change"]   = (df["pq_label"] != df["pq_label"].shift(1)).astype(int)
roll_instability     = df["order_change"].rolling(90).sum()

fig, axes = plt.subplots(4, 1, figsize=(16, 16))
fig.suptitle("MRF -- Rolling ARIMA Order Drift Analysis", fontsize=13)

order_map = {"(0,0)":0,"(1,0)":1,"(0,1)":2,"(1,1)":3}
colors_map = {0:"gray",1:"blue",2:"green",3:"red"}
for label, code in order_map.items():
    mask = df["pq_label"] == label
    axes[0].scatter(df[mask].index, [code]*mask.sum(), s=4,
                    color=colors_map[code],
                    label=f"ARIMA{label}")
axes[0].set_yticks([0,1,2,3])
axes[0].set_yticklabels(["(0,0)","(1,0)","(0,1)","(1,1)"])
axes[0].set_title("Selected ARIMA(p,q) Over Time"); axes[0].legend(fontsize=8, ncol=4)

df["AIC"].plot(ax=axes[1], color="navy", lw=0.8)
axes[1].set_title("Rolling AIC"); axes[1].set_ylabel("AIC")
```

```
df["ar_coef"].plot(ax=axes[2], color="green", lw=1)
axes[2].axhline(0, color="black", lw=0.8, linestyle="--")
axes[2].set_title("AR(1) Coefficient When Selected")

roll_instability.plot(ax=axes[3], color="red", lw=1.2)
axes[3].set_title("Rolling 90-Day Model Instability (order changes per 90d)")
plt.tight_layout(); plt.show()

print("Order Selection Frequency:")
print(df["pq_label"].value_counts())
print(f"\nMost common: ARIMA{df['pq_label'].mode()[0]}")
print(f"Peak instability: {roll_instability.idxmax().strftime('%Y-%m-%d')} "
      f"({roll_instability.max():.0f} changes in 90 days)")
```

② Plain English

Model drift analysis asks: "Does the best ARIMA order for MRF stay constant over time, or does it shift as market dynamics change?" High instability periods (many order changes in 90 days) indicate that no single ARIMA model is stable — a sign that the market is in transition or that ARIMA is misspecified for that period.

③ Common Mistake

Interpreting frequent order changes as evidence that ARIMA is "wrong" for MRF. It's actually evidence that the true model is time-varying — which is a finding in itself. The correct response is either to use a rolling window (which this code does), or switch to a regime-switching model that explicitly allows parameters to change over time.

④ Real World Link

Model stability monitoring is standard in quantitative risk management. If the ARIMA model selected for MRF returns switches multiple times in a month, the risk team would flag this as a "model instability event" and increase the volatility buffer until the model stabilises. Model drift detection is a regulatory requirement under BCBS model risk management guidelines.

Section C — Hard Problems — M5.016 to M5.023

M5.016 *Write from Scratch* HARD

Topic: BTC — Complete Quantitative Research Note

The Brief: Write a full research pipeline that produces a 5-section quantitative research note on BTC, ready to present to an investment committee.

Required sections: (1) Data Summary — descriptive statistics, fat tail confirmation; (2) Stationarity Analysis — ADF + KPSS; (3) Mean Model — auto-select ARIMA,

validate residuals; (4) Volatility Model — GARCH(1,1) with t-dist, risk metrics; (5) Forward View — 22-day return and volatility forecast with CI.

Solution

① Complete Code

```
!pip install yfinance statsmodels arch pmdarima -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, scipy.stats as spstats
import warnings; warnings.filterwarnings('ignore')
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.stats.diagnostic import acorr_ljungbox
from statsmodels.graphics.tsaplots import plot_acf
from arch import arch_model
from arch.univariate import arch_lm_test
from pmdarima import auto_arima
from statsmodels.tsa.arima.model import ARIMA

print("BTC-USD -- QUANTITATIVE RESEARCH NOTE"); print("="*60)

# Section 1: Data Summary
btc     = yf.download("BTC-USD", start="2019-01-01", end="2024-12-31", progress=False)
close   = btc["Close"].dropna()
returns = np.log(close).diff().dropna() * 100

n = len(returns); mu = returns.mean(); sigma = returns.std()
skew = spstats.skew(returns); kurt = spstats.kurtosis(returns)
jb_p = spstats.jarque_bera(returns).pvalue
roll_max = close.cummax(); max_dd = ((close-roll_max)/roll_max).min()

print(f"\n1. DATA SUMMARY")
print(f"  Observations    : {n}")
print(f"  Ann. Return     : {mu*252:.1f}%  Ann. Vol: {sigma*np.sqrt(252):.1f}%")
print(f"  Sharpe          : {(mu*252)/(sigma*np.sqrt(252)):.4f}")
print(f"  Skewness / ExKurt: {skew:.4f} / {kurt:.4f}")
print(f"  JB p-value      : {jb_p:.6f} ({'Non-normal' if jb_p<0.05 else 'Normal'})")
print(f"  Max Drawdown    : {max_dd*100:.1f}%")
print(f"  95% HS VaR      : {np.percentile(returns,5):.4f}%")

```

```
# Section 2: Stationarity
print(f"\n2. STATIONARITY")
adf = adfuller(returns); kpss_r = kpss(returns, regression="c", nlags="auto")
kpss_rej = kpss_r[0] > kpss_r[3]["5%"]
print(f"  ADF p={adf[1]:.6f}  KPSS: {'Non-stationary' if kpss_rej else 'Stationary'}")
print(f"  Both agree (stationary): {(adf[1]<0.05) and not kpss_rej}")
print(f"  -> d=0 for ARIMA (returns already stationary)")

# Section 3: Mean Model
print(f"\n3. MEAN MODEL")
auto_fit   = auto_arima(returns, start_p=0, max_p=2, start_q=0, max_q=2,
                        d=0, information_criterion="aic", stepwise=True,
                        suppress_warnings=True, error_action="ignore")
best_order = auto_fit.order
print(f"  Selected: ARIMA{best_order} (AIC={auto_fit.aic():.2f})")
arima_fit  = ARIMA(returns, order=best_order).fit()
lb_r  = acorr_ljungbox(arima_fit.resid.dropna(), lags=20, return_df=True)
lm_r  = arch_lm_test(arima_fit.resid.dropna(), nlags=10)
print(f"  Residual LB min p: {lb_r['lb_pvalue'].min():.4f} "
      f"({'OK' if lb_r['lb_pvalue'].min()>0.05 else 'Autocorrelation'})")
print(f"  Residual ARCH p  : {lm_r.pvalue:.4f} "
      f"({'GARCH needed' if lm_r.pvalue<0.05 else 'No ARCH'})")

# Section 4: Volatility Model
print(f"\n4. VOLATILITY MODEL")
garch_fit = arch_model(returns, mean="Constant", vol="GARCH",
                       p=1, q=1, dist="t").fit(disp="off")
alpha = garch_fit.params["alpha[1]"]; beta = garch_fit.params["beta[1]"]
omega = garch_fit.params["omega"];    nu   = garch_fit.params["nu"]
perst = alpha+beta; hl = np.log(0.5)/np.log(perst)
lrv   = omega/(1-perst); lrvol = np.sqrt(lrv)
latest_vol = garch_fit.conditional_volatility.iloc[-1]

print(f"  alpha={alpha:.4f} beta={beta:.4f} nu={nu:.2f}")
print(f"  Persistence: {perst:.4f}  Half-life: {hl:.1f}d")
```

```
print(f"  Long-run annual vol: {lrvol*np.sqrt(252):.2f}%")
print(f"  Current annual vol : {latest_vol*np.sqrt(252):.2f}%")
lm_std = arch_lm_test(garch_fit.std_resid.dropna(), nlags=10)
print(f"  Std resid ARCH p   : {lm_std.pvalue:.4f} "
      f"({'GARCH adequate' if lm_std.pvalue>0.05 else 'Needs higher order'})")

# Section 5: Forward View
print(f"\n5. FORWARD VIEW")
fc_mean = arima_fit.get_forecast(steps=22).predicted_mean.values
fc_vol  = np.sqrt(garch_fit.forecast(horizon=22, reindex=False).variance.iloc[0].values)
z95     = spstats.norm.ppf(0.975)

print(f"  22-day cumulative return: {np.sum(fc_mean):+.4f}%")
print(f"  Day-1: {fc_mean[0]:+.4f}% +/- {fc_vol[0]:.4f}%")
print(f"  Day-22 vol forecast: {fc_vol[-1]:.4f}%/day ({fc_vol[-1]*np.sqrt(252):.2f}%/yr)")

# Summary chart
fig, axes = plt.subplots(2,2, figsize=(18,12))
fig.suptitle("BTC-USD -- Quantitative Research Note", fontsize=14)
axes[0,0].hist(returns, bins=80, density=True, color="steelblue", alpha=0.6)
x = np.linspace(returns.min(), returns.max(), 300)
axes[0,0].plot(x, spstats.norm.pdf(x, mu, sigma), "r-", lw=2, label="Normal")
axes[0,0].plot(x, spstats.t.pdf(x, df=nu, loc=mu, scale=sigma), "g-", lw=2, label=f"t({nu:.1f})")
axes[0,0].set_title("Return Distribution vs Models"); axes[0,0].legend()
garch_fit.conditional_volatility.plot(ax=axes[0,1], color="red", lw=0.8)
axes[0,1].axhline(lrvol, color="black", lw=1.5, linestyle="--", label=f"LR vol={lrvol:.2f}%")
axes[0,1].set_title("GARCH Conditional Volatility"); axes[0,1].legend()
plot_acf(returns**2, lags=30, ax=axes[1,0], zero=False, color="orange")
axes[1,0].set_title("ACF Squared Returns -- ARCH Confirmation")
hist_t = range(60); fc_t = range(60,82)
axes[1,1].plot(hist_t, returns[-60:].values, color="steelblue", lw=0.8)
```

```
axes[1,1].plot(fc_t, fc_mean, "r-o", markersize=4, lw=1.5,
    label="22d Forecast")
axes[1,1].fill_between(fc_t, fc_mean-z95*fc_vol,
    fc_mean+z95*fc_vol,
                           color="red", alpha=0.2, label="95%
                              CI")
axes[1,1].axhline(0, color="black", lw=0.8, linestyle="--")
axes[1,1].set_title("22-Day Return Forecast");
    axes[1,1].legend(fontsize=8)
plt.tight_layout(); plt.show()
print("\n" + "="*60 + "\nEND OF BTC-USD QUANTITATIVE
    RESEARCH NOTE\n" + "="*60)
```

② Plain English

This research note structure — Data → Stationarity → Mean Model → Volatility Model → Forward View — is the standard template for quantitative equity/crypto research. Each section answers one fundamental question: What does the data look like? Is it safe to model directly? What is the expected return? How uncertain is that expectation? What do we forecast?

③ Common Mistake

Presenting a forward view without stating model limitations. A proper research note adds: "These forecasts assume GARCH(1,1) adequately captures BTC's variance dynamics. Model risk includes: (a) parameter instability during regime changes; (b) fat tails not fully captured even with t-distribution; (c) structural breaks near major events." Quantitative humility is a professional virtue.

④ Real World Link

Investment banks, crypto trading firms, and asset managers publish precisely this type of quantitative research note. Galaxy Digital, Ark Invest, and Grayscale all publish BTC quantitative analyses that include exactly these five sections. The difference between student work and professional output is mostly presentation polish and the "limitations" section — the methodology is identical.

M5.017 *Write from Scratch* **HARD**

Topic: MRF — Event Study: COVID Crash Impact Analysis

The Brief: Quantify exactly how COVID (March–April 2020) impacted MRF's return dynamics using a rigorous event study.

Write code covering: (1) define event window [-5, +60] days around March 16, 2020; (2) estimate normal returns using pre-event ARIMA (trained on 2019); (3) compute abnormal returns = actual − expected; (4) compute cumulative abnormal returns (CAR); (5) GARCH-based volatility before, during, and after event; (6) statistical test: are event-window returns significantly different from normal?

Solution

① Complete Code

```
!pip install yfinance statsmodels arch -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, scipy.stats as spstats
import warnings; warnings.filterwarnings('ignore')
from statsmodels.tsa.arima.model import ARIMA
from arch import arch_model

mrf       = yf.download("MRF.NS", start="2019-01-01",
   end="2021-06-30", progress=False)
returns   = np.log(mrf["Close"]).diff().dropna() * 100
event_date = pd.Timestamp("2020-03-16")
nearest_idx= returns.index.get_indexer([event_date],
   method="nearest")[0]
event_actual = returns.index[nearest_idx]
print(f"Event date: {event_actual.date()}")

pre_event = returns[:event_actual - pd.Timedelta(days=1)]
arima_pre = ARIMA(pre_event, order=(1,0,1)).fit()

idx_pos = list(returns.index).index(event_actual)
start_i = max(0, idx_pos-5); end_i = min(len(returns),
   idx_pos+61)
event_window = returns.iloc[start_i:end_i]; n_event =
   len(event_window)

normal_fc = arima_pre.forecast(steps=n_event).values
actual_v  = event_window.values
abnormal  = actual_v - normal_fc
car       = np.cumsum(abnormal)
t_stat, t_p = spstats.ttest_1samp(abnormal, 0)

garch_full = arch_model(returns, mean="Constant",
   vol="GARCH", p=1, q=1).fit(disp="off")
cond_vol   = garch_full.conditional_volatility
pre_vol    = cond_vol[:event_actual -
   pd.Timedelta(days=1)].mean()
event_vol  = cond_vol[event_window.index].mean()
post_vol   = cond_vol[event_actual +
   pd.Timedelta(days=61):].mean()

print(f"\n{'='*60}")
print(f"  MRF -- COVID Event Study:
   {event_actual.strftime('%b %d, %Y')}")
print(f"{'='*60}")
print(f"  Abnormal Returns: mean={abnormal.mean():+.4f}%
   CAR={car[-1]:+.4f}%")
```

```
print(f"  t-stat: {t_stat:.4f}  p={t_p:.4f}  Significant: {t_p<0.05}")
print(f"\n  GARCH Volatility:")
print(f"    Pre-event : {pre_vol:.4f}%/day ({pre_vol*np.sqrt(252):.1f}% p.a.)")
print(f"    Event     : {event_vol:.4f}%/day ({event_vol*np.sqrt(252):.1f}% p.a.)")
print(f"    Post-event: {post_vol:.4f}%/day ({post_vol*np.sqrt(252):.1f}% p.a.)")
print(f"    Vol spike : {event_vol/pre_vol:.2f}x  Recovery: {post_vol/pre_vol:.2f}x")

days = range(n_event)
fig, axes = plt.subplots(3,1, figsize=(16,14))
axes[0].bar(days, actual_v, color="steelblue", alpha=0.7, label="Actual Returns")
axes[0].plot(days, normal_fc, "r--", lw=2, label="ARIMA Normal Returns")
axes[0].axvline(5, color="black", lw=2, linestyle=":", label="Event Day")
axes[0].set_title("MRF -- Actual vs Expected Returns"); axes[0].legend()

axes[1].fill_between(days, car, 0, where=(car>=0), color="green", alpha=0.4)
axes[1].fill_between(days, car, 0, where=(car<0), color="red",   alpha=0.4)
axes[1].plot(days, car, "k-", lw=2)
axes[1].axvline(5, color="black", lw=2, linestyle=":")
axes[1].axhline(0, color="gray", lw=0.8)
axes[1].set_title(f"Cumulative Abnormal Return -- Total CAR: {car[-1]:+.2f}%")

vol_window = cond_vol[event_window.index]
axes[2].plot(range(n_event), vol_window.values, color="red", lw=1.5)
axes[2].axhline(pre_vol, color="green", lw=1.5, linestyle="--",
                label=f"Pre-event avg: {pre_vol:.2f}%")
axes[2].axvline(5, color="black", lw=2, linestyle=":")
axes[2].set_title("GARCH Conditional Volatility During Event")
axes[2].legend(); axes[2].set_xlabel("Trading Days from Event")
plt.tight_layout(); plt.show()
```

② Plain English

An event study separates how much of the return around COVID was "expected" (the ARIMA model's normal forecast) vs "abnormal" (the unexpected shock from

the event). For MRF, the COVID crash likely produced strongly negative cumulative abnormal returns — returns that couldn't be predicted from the pre-event ARIMA model trained only on 2019 data.

③ Common Mistake

Using the full-period ARIMA (trained on 2019–2024 data) to estimate "normal" returns during 2020. This introduces look-ahead bias — the model "knows" what 2020 returns look like. The correct approach trains the ARIMA exclusively on pre-event data (here: only 2019), so the normal forecast represents genuine pre-event expectations.

④ Real World Link

Event studies are the standard methodology in academic finance for measuring market impact. Every major paper on "how does X affect stock prices?" uses this framework. The CAR statistic is the primary reported result in academic papers studying event impacts on equity prices.

M5.018 *Write from Scratch* HARD

Topic: Gold — Complete Factor Decomposition

The Brief: Decompose Gold's return into: (1) global macro component (modelled by BTC as a risk-on/off proxy), (2) seasonal component, (3) GARCH volatility component, and (4) idiosyncratic residual.

Write complete code implementing this 4-component decomposition and quantify how much variance each component explains.

Solution

① Complete Code

```
!pip install yfinance statsmodels arch -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.seasonal import seasonal_decompose
from arch import arch_model

gold = yf.download("GC=F",    start="2019-01-01",
   end="2024-12-31", progress=False)
btc  = yf.download("BTC-USD", start="2019-01-01",
   end="2024-12-31", progress=False)

gold_ret = np.log(gold["Close"]).diff().dropna() * 100
btc_ret  = np.log(btc["Close"]).diff().dropna() * 100
combined =
   gold_ret.to_frame("Gold").join(btc_ret.to_frame("BTC"),how="inner").d
print(f"Aligned: {len(combined)} observations")
```

```
16
17 # Component 1: Macro Factor (BTC influence via SARIMAX)
18 sarimax_fit  = SARIMAX(combined["Gold"],
      exog=combined["BTC"],
19                        order=(1,0,1)).fit(disp=False)
20 btc_coef     = sarimax_fit.params["BTC"]
21 macro_comp   = sarimax_fit.fittedvalues
22 arima_resid  = sarimax_fit.resid
23
24 print(f"\nMacro Component:")
25 print(f"  BTC coefficient: {btc_coef:.6f}")
26 print(f"  BTC p-value    :
      {sarimax_fit.pvalues['BTC']:.4f} "
27       f"({'Significant' if sarimax_fit.pvalues['BTC']<0.05
            else 'Not significant'})")
28
29 # Component 2: Seasonal (from monthly decomposition)
30 gold_monthly = gold["Close"].dropna().resample("ME").last()
31 decomp = seasonal_decompose(gold_monthly,
      model="multiplicative", period=12)
32 seas_monthly = decomp.seasonal
33 seasonal_daily = pd.Series(index=combined.index,
      dtype=float)
34 for date in combined.index:
35     month_end = pd.Timestamp(date.year, date.month, 1) +
          pd.offsets.MonthEnd(0)
36     seasonal_daily[date] = seas_monthly[month_end] if
          month_end in seas_monthly.index else 1.0
37 seas_contrib = (seasonal_daily - 1) * 100
38
39 # Component 3: GARCH volatility
40 garch = arch_model(combined["Gold"], mean="Constant",
41                    vol="GARCH", p=1, q=1).fit(disp="off")
42 vol_comp    = garch.conditional_volatility
43 idiosync    = garch.std_resid.dropna()
44
45 # Variance decomposition
46 r2_macro = 1 - arima_resid.var() / combined["Gold"].var()
47 r2_arch  = 1 - (idiosync.std()**2 / combined["Gold"].var())
48
49 print(f"\nVariance Decomposition:")
50 print(f"  R2 from SARIMAX (macro+ARMA): {r2_macro:.4f}
      ({r2_macro*100:.1f}%)")
51 print(f"  R2 from GARCH (vol)          : {r2_arch:.4f}
      ({r2_arch*100:.1f}%)")
52 print(f"  Idiosyncratic residual        :
      {(1-r2_arch)*100:.1f}%")
53
54 fig, axes = plt.subplots(4,1, figsize=(16,18))
```

```
fig.suptitle("Gold Returns -- 4-Component Decomposition",
    fontsize=13)
combined["Gold"].plot(ax=axes[0], color="gold",   lw=0.8)
axes[0].set_title("Total Gold Return (Original)")
macro_comp.plot(ax=axes[1],       color="blue",   lw=1)
axes[1].axhline(0, color="gray", lw=0.8, linestyle="--")
axes[1].set_title("Component 1: Macro Factor (SARIMAX with
    BTC exogenous)")
seas_contrib.plot(ax=axes[2],     color="green",  lw=1)
axes[2].axhline(0, color="gray", lw=0.8, linestyle="--")
axes[2].set_title("Component 2: Seasonal Pattern")
vol_comp.reindex(combined.index).plot(ax=axes[3],
    color="red", lw=0.8)
axes[3].set_title("Component 3: GARCH Conditional
    Volatility")
plt.tight_layout(); plt.show()
```

② Plain English

This decomposition is a simplified version of factor models used in professional asset management. By attributing Gold's return to macro/risk-appetite (via BTC), seasonal demand patterns, and GARCH volatility dynamics, we identify what is "explainable" vs "idiosyncratic." The idiosyncratic residual represents genuine Gold-specific information (mine supply shocks, geopolitical events) that none of the systematic factors captured.

③ Common Mistake

Treating the variance decomposition as precise percentages that sum to 100%. The components are not orthogonal — macro, seasonal, and GARCH components may be correlated, so their variance contributions don't cleanly partition the total. The decomposition is approximate and indicative, not exact.

④ Real World Link

Professional commodity research firms publish exactly this kind of decomposition for Gold. Their factor models typically include: DXY (dollar index), real interest rates, risk appetite (VIX or equity returns), seasonal demand, and central bank buying — each quantified as a coefficient. The "unexplained" idiosyncratic component is what makes trading interesting.

M5.019 *Write from Scratch* **HARD**

Topic: Three-Asset Portfolio — Mean-Variance with GARCH Covariance
The Brief: Build a mean-variance efficient portfolio using BTC, MRF, and Gold, but replace the static historical covariance matrix with a GARCH-based dynamic covariance matrix. Compare the efficient frontier under both assumptions.

Solution

① Complete Code

```
!pip install yfinance arch statsmodels scipy -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import minimize
import warnings; warnings.filterwarnings('ignore')
from arch import arch_model

tickers = {"BTC":"BTC-USD","MRF":"MRF.NS","Gold":"GC=F"}
ret_dict = {}
for name, ticker in tickers.items():
    raw = yf.download(ticker, start="2020-01-01",
       end="2024-12-31", progress=False)
    ret_dict[name] = np.log(raw["Close"]).diff().dropna()
       * 100

returns  = pd.DataFrame(ret_dict).dropna()
n_assets = 3; assets = list(returns.columns)

# Static covariance
mu_static  = returns.mean().values * 252
cov_static = returns.cov().values  * 252

# GARCH-based dynamic covariance (DCC EWMA)
garch_fits = {}; std_resids = {}
for asset in assets:
    fit = arch_model(returns[asset], mean="Constant",
                     vol="GARCH", p=1, q=1).fit(disp="off")
    garch_fits[asset] = fit; std_resids[asset] =
       fit.std_resid.dropna()

std_df = pd.DataFrame(std_resids).dropna()
lam = 0.94; Q_t = np.cov(std_df.values.T)
for t in range(1, len(std_df)):
    z = std_df.iloc[t-1].values.reshape(-1,1)
    Q_t = lam*Q_t + (1-lam)*(z @ z.T)

D_inv = np.diag(1/np.sqrt(np.diag(Q_t)))
dcc_corr = D_inv @ Q_t @ D_inv
garch_vols =
   np.array([garch_fits[a].conditional_volatility.iloc[-1]
                        for a in assets]) * np.sqrt(252)
D = np.diag(garch_vols)
cov_garch = D @ dcc_corr @ D

print("Latest DCC Correlation:")
print(pd.DataFrame(dcc_corr, index=assets,
   columns=assets).round(4))
```

```
print(f"\nGARCH annual vols: {dict(zip(assets, garch_vols.round(2)))}")

def portfolio_stats(w, mu, cov):
    r = w@mu; v = np.sqrt(w@cov@w); return r, v, r/v if v>0 else 0

def max_sharpe(mu, cov):
    n = len(mu)
    def neg_sr(w): return -portfolio_stats(w, mu, cov)[2]
    r = minimize(neg_sr, np.ones(n)/n, bounds=[(0,1)]*n,
                 constraints=[{"type":"eq","fun":lambda w: w.sum()-1}], method="SLSQP")
    return r.x if r.success else np.ones(n)/n

w_static = max_sharpe(mu_static, cov_static)
w_garch  = max_sharpe(mu_static, cov_garch)
rs, vs, shs = portfolio_stats(w_static, mu_static, cov_static)
rg, vg, shg = portfolio_stats(w_garch,  mu_static, cov_garch)

def frontier(mu, cov):
    pts = []
    for tr in np.linspace(mu.min()*0.5, mu.max()*1.2, 60):
        def obj(w): return np.sqrt(w@cov@w)
        r = minimize(obj, np.ones(len(mu))/len(mu),
            bounds=[(0,1)]*len(mu),
                     constraints=[{"type":"eq","fun":lambda w: w.sum()-1},
                                  {"type":"eq","fun":lambda w: w@mu-tr}],
                                  method="SLSQP")
        if r.success: pts.append((np.sqrt(r.x@cov@r.x), r.x@mu))
    return zip(*pts) if pts else ([],[])

fig, axes = plt.subplots(1,2, figsize=(18,7))
vs_f, rs_f = frontier(mu_static, cov_static)
vg_f, rg_f = frontier(mu_static, cov_garch)
axes[0].plot(list(vs_f), list(rs_f), "b-", lw=2, label="Static Frontier")
axes[0].plot(list(vg_f), list(rg_f), "r-", lw=2, label="GARCH Frontier")
axes[0].scatter([vs],[rs], color="blue", s=100, marker="*", zorder=5,
                label=f"Static MaxSharpe (SR={shs:.2f})")
axes[0].scatter([vg],[rg], color="red",  s=100, marker="*", zorder=5,
```

```
                  label=f"GARCH MaxSharpe (SR={shg:.2f})")
for i, a in enumerate(assets):
    axes[0].scatter(np.sqrt(cov_static[i,i]),
       mu_static[i], s=80, zorder=5)
    axes[0].annotate(a, (np.sqrt(cov_static[i,i]),
       mu_static[i]),
                     textcoords="offset points",
                        xytext=(5,3), fontsize=10)
axes[0].set_xlabel("Annual Vol (%)");
   axes[0].set_ylabel("Annual Return (%)")
axes[0].set_title("Efficient Frontier: Static vs GARCH");
   axes[0].legend(fontsize=9)

bar_w = 0.35; x = np.arange(n_assets)
axes[1].bar(x-bar_w/2, w_static*100, bar_w, color="blue",
   alpha=0.7, label="Static")
axes[1].bar(x+bar_w/2, w_garch*100,  bar_w, color="red",
   alpha=0.7, label="GARCH")
axes[1].set_xticks(x); axes[1].set_xticklabels(assets,
   fontsize=12)
axes[1].set_title("Max Sharpe Weights");
   axes[1].set_ylabel("Weight (%)"); axes[1].legend()

plt.suptitle("BTC-MRF-Gold Portfolio: Static vs GARCH
   Covariance", fontsize=13)
plt.tight_layout(); plt.show()

print(f"\nMax Sharpe Portfolios:")
for i, a in enumerate(assets):
    print(f"  {a}: Static={w_static[i]*100:.1f}%
       GARCH={w_garch[i]*100:.1f}%")
```

② Plain English

When you replace the static historical covariance matrix with the GARCH-based time-varying covariance, the efficient frontier shifts. Assets currently in high-volatility regimes look less attractive (their risk is elevated), so the optimiser moves weight toward currently-calmer assets. This is dynamic risk management — the portfolio automatically responds to changing market conditions.

③ Common Mistake

Using the full-period historical mean returns for GARCH-based optimisation but dynamic covariance. This creates an inconsistency: the return estimates are static but the risk estimates are dynamic. In practice, both should be dynamic, or return forecasts should come from ARIMA models fitted to the same period.

④ Real World Link

BlackRock's Aladdin risk system, JPMorgan's risk models, and virtually every major asset manager uses dynamic covariance estimation rather than static historical covariance. The improvement in Sharpe ratio from GARCH vs static covariance is

typically 0.05–0.20 — modest but consistent, which compounds significantly over time.

M5.020 *Write from Scratch* **HARD**

Topic: BTC-MRF-Gold — Granger Causality Network Visualisation
The Brief: Build a directional causal network showing which assets lead which at different frequencies (daily vs weekly), and visualise it as an annotated network diagram with arrows showing significant causal links.

Solution

① **Complete Code**

```
!pip install yfinance statsmodels -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from statsmodels.tsa.vector_ar.var_model import VAR

def build_causality_network(data, max_lags=5):
    assets = list(data.columns)
    fit    = VAR(data).fit(maxlags=max_lags, ic="aic")
    p_mat  = {}
    for caused in assets:
        p_mat[caused] = {}
        for causing in assets:
            if caused == causing: p_mat[caused][causing] =
               np.nan
            else:
                gc = fit.test_causality(caused=caused,
                   causing=[causing])
                p_mat[caused][causing] = gc.pvalue
    return pd.DataFrame(p_mat).T, fit.k_ar

tickers = {"BTC":"BTC-USD","MRF":"MRF.NS","Gold":"GC=F"}
ret_dict = {}
for name, ticker in tickers.items():
    raw = yf.download(ticker, start="2020-01-01",
       end="2024-12-31", progress=False)
    ret_dict[name] = np.log(raw["Close"]).diff().dropna()
       * 100

daily_data  = pd.DataFrame(ret_dict).dropna()
weekly_data = daily_data.resample("W").sum().dropna()

daily_gc,  d_lags = build_causality_network(daily_data)
weekly_gc, w_lags = build_causality_network(weekly_data)

```

```
print(f"Daily VAR({d_lags}) Granger p-values:")
print(daily_gc.round(4)); print(f"\nWeekly VAR({w_lags}) Granger p-values:")
print(weekly_gc.round(4))

def draw_network(ax, gc_matrix, title, positions, threshold=0.10):
    assets = gc_matrix.index.tolist()
    ax.set_xlim(-0.2,1.2); ax.set_ylim(-0.2,1.2)
    ax.set_aspect("equal"); ax.axis("off"); ax.set_title(title, fontsize=13)
    nc = {"BTC":"#F7931A","MRF":"#1E90FF","Gold":"#FFD700"}
    for asset, (x,y) in positions.items():
        ax.add_patch(plt.Circle((x,y), 0.12, color=nc[asset], zorder=5))
        ax.text(x, y, asset, ha="center", va="center", fontsize=12,
                fontweight="bold", zorder=6,
                color="white" if asset=="BTC" else "black")
    for causing in assets:
        for caused in assets:
            if causing == caused: continue
            p = gc_matrix.loc[causing, caused]
            if pd.isna(p) or p >= threshold: continue
            x1,y1 = positions[causing]; x2,y2 = positions[caused]
            dx,dy = x2-x1, y2-y1; dist = np.sqrt(dx**2+dy**2)
            ux,uy = dx/dist, dy/dist
            sx = x1+ux*0.13; sy = y1+uy*0.13
            ex = x2-ux*0.13; ey = y2-uy*0.13
            lw = 3 if p<0.01 else 1.5
            col = "darkred" if p<0.01 else "darkorange"
            ax.annotate("", xy=(ex,ey), xytext=(sx,sy),
                        arrowprops=dict(arrowstyle="-|>", color=col,
                                        lw=lw, mutation_scale=20))
            mx = (sx+ex)/2+uy*0.06; my = (sy+ey)/2-ux*0.06
            ax.text(mx, my, f"p={p:.3f}", fontsize=7, ha="center", va="center",
                    color=col, bbox=dict(boxstyle="round,pad=0.15",
                    facecolor="white", alpha=0.8))

positions = {"BTC":(0.5,0.85),"MRF":(0.1,0.15),"Gold":(0.9,0.15)}
fig, axes = plt.subplots(1,2, figsize=(18,9))
```

```
fig.suptitle("Granger Causality Networks: BTC - MRF - Gold\n"
             "Arrow = significant lagged predictive relationship", fontsize=13)
draw_network(axes[0], daily_gc,  f"Daily Frequency (VAR-{d_lags})", positions)
draw_network(axes[1], weekly_gc, f"Weekly Frequency (VAR-{w_lags})", positions)
plt.tight_layout(); plt.show()

print("\nSignificant causal links (p<0.10):")
for freq_name, gc_mat in [("Daily",daily_gc),("Weekly",weekly_gc)]:
    print(f"\n  {freq_name}:")
    for causing in gc_mat.index:
        for caused in gc_mat.columns:
            if causing != caused:
                p = gc_mat.loc[causing,caused]
                if not pd.isna(p) and p < 0.10:
                    print(f"    {causing} -> {caused}: p={p:.4f} "
                          f"({'**' if p<0.01 else '*'})")
```

② Plain English

The Granger causality network shows which assets have predictive power over which others. An arrow from BTC to Gold means: after controlling for Gold's own history, BTC's past returns add predictive power for Gold's future returns. Arrows in both directions indicate bidirectional lead-lag relationships. The comparison between daily and weekly frequencies reveals whether cross-asset dynamics operate over short or longer horizons.

③ Common Mistake

Interpreting Granger causality arrows as "BTC prices move Gold prices." Granger causality is purely statistical and says "BTC's past returns contain predictive information about Gold's future returns." This doesn't establish economic causation — both might be driven by a common third factor.

④ Real World Link

Cross-asset Granger causality networks are used by macro trading desks to understand information flow across markets. If BTC reliably leads Gold by 2–3 days at the weekly frequency, a macro trader can use a large BTC move as an early signal for a Gold position. This "lead-lag exploitation" is one of the legitimate sources of alpha in systematic macro trading.

M5.021 *Write from Scratch* **HARD**

Topic: MRF — Kalman Filter vs ARIMA for Trend Extraction

The Brief: Compare three trend extraction methods for MRF price: HP filter (via

statsmodels), seasonal decomposition trend, and Holt's double exponential smoothing. Then use each extracted trend to build a trend-following signal and compare performance.

Solution

① Complete Code

```
!pip install yfinance statsmodels -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, warnings
warnings.filterwarnings('ignore')
from statsmodels.tsa.filters.hp_filter import hpfilter
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.holtwinters import
   ExponentialSmoothing

mrf    = yf.download("MRF.NS", start="2019-01-01",
   end="2024-12-31", progress=False)
close = mrf["Close"].dropna()

# Method 1: Hodrick-Prescott Filter
lambda_hp  = 1600 * (252**2)    # Ravn-Uhlig rule for daily
   data
cycle_hp, trend_hp = hpfilter(close, lamb=lambda_hp)

# Method 2: Seasonal Decomposition Trend (weekly then
   interpolate to daily)
weekly      = close.resample("W").last()
decomp      = seasonal_decompose(weekly,
   model="multiplicative", period=52)
trend_decomp=
   decomp.trend.dropna().reindex(close.index).interpolate(method="time")

# Method 3: Holt's Double Exponential Smoothing
hw_fit     = ExponentialSmoothing(close, trend="add",
   damped_trend=True).fit(optimized=True)
trend_hw   = hw_fit.fittedvalues

def trend_strategy(price, trend, name):
    signal      = pd.Series(np.where(price > trend, 1, -1),
       index=price.index).shift(1)
    ret         = np.log(price).diff().dropna()
    strat_ret   = signal.reindex(ret.index) * ret
    bh_ret      = ret
    cum_strat   = (1 + strat_ret.dropna()).cumprod()
    cum_bh      = (1 + bh_ret.dropna()).cumprod()
    sh_strat    = (strat_ret.mean()/strat_ret.std()) *
       np.sqrt(252)
```

```
    sh_bh       = (bh_ret.mean()/bh_ret.std())        *
       np.sqrt(252)
    return cum_strat, cum_bh, sh_strat, sh_bh

cs_hp, cb, sh_hp, sh_bh = trend_strategy(close, trend_hp,
       "HP")
cs_dc, _,  sh_dc, _     = trend_strategy(close,
   trend_decomp, "Decomp")
cs_hw, _,  sh_hw, _     = trend_strategy(close, trend_hw,
       "Holts")

fig, axes = plt.subplots(2,2, figsize=(18,12))
fig.suptitle("MRF -- Trend Extraction Comparison",
   fontsize=13)
axes[0,0].plot(close,         color="gray",   lw=0.8,
   alpha=0.6, label="MRF Price")
axes[0,0].plot(trend_hp,      color="blue",   lw=2,
   label="HP Filter")
axes[0,0].plot(trend_decomp, color="green",  lw=2,
   label="Decomposition")
axes[0,0].plot(trend_hw,      color="red",    lw=2,
   label="Holt's")
axes[0,0].set_title("Price + All Three Trend Estimates");
   axes[0,0].legend(fontsize=8)

cycle_hp.plot(ax=axes[0,1], color="purple", lw=0.8)
axes[0,1].axhline(0, color="black", lw=1, linestyle="--")
axes[0,1].set_title("HP Filter Cycle Component
   (Detrended)")

axes[1,0].plot(cs_hp, color="blue",  lw=1.5, label=f"HP
   (Sharpe={sh_hp:.2f})")
axes[1,0].plot(cs_dc, color="green", lw=1.5,
   label=f"Decomp (Sharpe={sh_dc:.2f})")
axes[1,0].plot(cs_hw, color="red",   lw=1.5, label=f"Holts
   (Sharpe={sh_hw:.2f})")
axes[1,0].plot(cb,    color="black", lw=2, linestyle="--",
               label=f"Buy-Hold (Sharpe={sh_bh:.2f})")
axes[1,0].set_title("Cumulative Return -- Trend-Following
   Strategies")
axes[1,0].legend(fontsize=8)

for col, (spt, st, name, color) in enumerate([
        (close, trend_hp,     "HP",    "blue"),
        (close, trend_decomp, "Decomp","green"),
        (close, trend_hw,     "Holts", "red")]):
    sig = pd.Series((spt>st).astype(int)*2-1,
       index=close.index)
```

```
    axes[1,1].plot(sig.rolling(90).mean(), color=color,
        lw=1.2, label=name)
axes[1,1].axhline(0, color="gray", lw=0.8, linestyle="--")
axes[1,1].set_title("Rolling 90-Day Bullish Signal");
    axes[1,1].legend(fontsize=9)
plt.tight_layout(); plt.show()

print("\nTrend-Following Strategy Sharpe Ratios:")
for name, sh in [("HP
    Filter",sh_hp),("Decomposition",sh_dc),("Holts",sh_hw)]:
     print(f"  {name:15} Sharpe={sh:.4f}  vs Buy-Hold:
        {sh-sh_bh:+.4f}")
print(f"  Buy-Hold        Sharpe={sh_bh:.4f}  (benchmark)")
best =
    max([("HP",sh_hp),("Decomp",sh_dc),("Holts",sh_hw)],
    key=lambda x:x[1])
print(f"\nBest trend estimator: {best[0]}
    (Sharpe={best[1]:.4f})")
```

② Plain English

Different trend extractors make different assumptions. HP filter minimises a combination of fit and smoothness. Seasonal decomposition extracts the trend as what's left after removing seasonality and residuals. Holt's double exponential smoothing is adaptive — it learns the trend speed from data. The "best" one depends on whether you want smoothness (HP) or responsiveness (Holt's) in your trading signal.

③ Common Mistake

Using the default HP filter $\lambda = 1600$ (designed for quarterly macroeconomic data) on daily stock data. With daily data, you need $\lambda \approx 1600 \times (252/4)^2 \approx 1.6$ million. Using $\lambda = 1600$ on daily data produces a trend that follows the price almost exactly (no smoothing), making it useless for signal generation.

④ Real World Link

Trend-following (also called momentum or CTA strategy) is one of the most extensively documented quantitative strategy styles. Winton Group, Man AHL, and Campbell & Company all run trend-following strategies on Indian equities including MRF. The core idea — buy above trend, sell below trend — is identical to this simplified version.

M5.022 *Write from Scratch* **HARD**

Topic: BTC — Tail Risk Comparison: Historical vs GARCH-EVT

The Brief: Compare three methods for estimating BTC's extreme tail risk (99.9% VaR and Expected Shortfall): (1) Historical Simulation, (2) GARCH-Normal, (3) GARCH with Extreme Value Theory (EVT) tail fitting using GPD.

Solution

① Complete Code

```
!pip install yfinance arch statsmodels scipy -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, scipy.stats as spstats
import warnings; warnings.filterwarnings('ignore')
from arch import arch_model

btc     = yf.download("BTC-USD", start="2019-01-01",
   end="2024-12-31", progress=False)
returns = np.log(btc["Close"]).diff().dropna() * 100
n = len(returns); alpha = 0.001    # 99.9%

# Method 1: Historical Simulation
hs_var = np.percentile(returns, alpha*100)
hs_es  = returns[returns < hs_var].mean()

# Method 2: GARCH-Normal
garch_n  = arch_model(returns, mean="Constant",
   vol="GARCH", p=1, q=1,
                      dist="normal").fit(disp="off")
lv_n     = garch_n.conditional_volatility.iloc[-1]; mu_n =
   garch_n.params["mu"]
q_norm   = spstats.norm.ppf(alpha)
gn_var   = mu_n + q_norm * lv_n
gn_es    = mu_n + (-spstats.norm.pdf(q_norm)/alpha) * lv_n

# Method 3: GARCH-EVT (GPD on standardised residuals)
garch_t  = arch_model(returns, mean="Constant",
   vol="GARCH", p=1, q=1,
                      dist="t").fit(disp="off")
std_resid = garch_t.std_resid.dropna().values
lv_t      = garch_t.conditional_volatility.iloc[-1]; mu_t
   = garch_t.params["mu"]

u_threshold = np.percentile(std_resid, 10)
tail_exc    = -(std_resid[std_resid < u_threshold] -
   u_threshold)
shape, loc, scale = spstats.genpareto.fit(tail_exc, floc=0)

N_u = (std_resid < u_threshold).sum()
var_evt_std = -(u_threshold +
   scale/shape*((n/N_u*alpha)**(-shape)-1))
es_evt_std  = var_evt_std/(1-shape) +
   (scale-shape*u_threshold)/(1-shape)
gevt_var    = mu_t + var_evt_std * lv_t
gevt_es     = mu_t + es_evt_std  * lv_t

print(f"{'='*60}")
```

```
print(f"  BTC-USD -- 99.9% VaR and ES Comparison")
print(f"{'='*60}")
print(f"  {'Method':25} {'99.9% VaR':>12} {'99.9% ES':>12}")
print(f"  {'-'*52}")
print(f"  {'Historical Simulation':25} {hs_var:>12.4f}% {hs_es:>12.4f}%")
print(f"  {'GARCH-Normal':25} {gn_var:>12.4f}% {gn_es:>12.4f}%")
print(f"  {'GARCH-EVT':25} {gevt_var:>12.4f}% {gevt_es:>12.4f}%")
print(f"{'='*60}")
print(f"\nGPD shape xi = {shape:.4f} "
      f"({'Heavy tail -- EVT more conservative' if shape>0
         else 'Bounded tail'})")

fig, axes = plt.subplots(1,2, figsize=(16,6))
fig.suptitle("BTC-USD -- 99.9% Tail Risk: 3 Methods", fontsize=13)
axes[0].hist(returns, bins=100, density=True, color="steelblue", alpha=0.5)
axes[0].axvline(hs_var,   color="blue",   lw=2, linestyle="--", label=f"HS: {hs_var:.2f}%")
axes[0].axvline(gn_var,   color="orange", lw=2, linestyle="--", label=f"GARCH-N: {gn_var:.2f}%")
axes[0].axvline(gevt_var, color="red",    lw=2, label=f"GARCH-EVT: {gevt_var:.2f}%")
axes[0].set_title("Return Distribution + 99.9% VaR Thresholds")
axes[0].legend(fontsize=8);
   axes[0].set_xlim(returns.min(), 5)

methods = ["Hist.Sim.","GARCH-N","GARCH-EVT"]
vars_   = [abs(hs_var), abs(gn_var), abs(gevt_var)]
ess_    = [abs(hs_es),  abs(gn_es),  abs(gevt_es)]
x_pos   = np.arange(3)
axes[1].bar(x_pos-0.2, vars_, 0.35, label="99.9% VaR", color="steelblue")
axes[1].bar(x_pos+0.2, ess_,  0.35, label="99.9% ES", color="red", alpha=0.7)
axes[1].set_xticks(x_pos); axes[1].set_xticklabels(methods)
axes[1].set_title("VaR and ES by Method"); axes[1].legend()
for i,(v,e) in enumerate(zip(vars_,ess_)):
    axes[1].text(i-0.2, v+0.05, f"{v:.2f}%", ha="center", fontsize=9)
    axes[1].text(i+0.2, e+0.05, f"{e:.2f}%", ha="center", fontsize=9)
plt.tight_layout(); plt.show()
```

② Plain English

The three methods make progressively more sophisticated tail assumptions. Historical Simulation uses raw past data — simple but doesn't adapt to current volatility. GARCH-Normal scales for current vol but underestimates tail thickness (fat tails). GARCH-EVT scales for current vol AND fits the tail's statistical shape using Extreme Value Theory — the most rigorous approach.

③ Common Mistake

Using Historical Simulation with a short lookback (1 year). If BTC's worst historical day was −15% in 2019 and you're computing VaR in a calm 2024 period using 1-year lookback, your HS-VaR will be around −3% (reflecting 2024's low volatility). GARCH-based VaR adapts to current vol but preserves tail shape information from the full history.

④ Real World Link

GARCH-EVT is the industry gold standard for tail risk in crypto. Major crypto exchanges use similar models to set margin requirements and liquidation thresholds. The key output — Expected Shortfall at 99.9% — directly determines how much capital a crypto exchange must hold as a buffer against extreme market moves.

M5.023 *Multiple Choice (Integrative)* **HARD**

Topic: Choosing the right model for the right question

You are given four research questions about MRF. Match each question to the **correct primary modelling approach**.

#	Research Question
1	"Will MRF's daily return tomorrow be positive or negative?"
2	"How large will MRF's price swings be over the next month?"
3	"Does a large BTC move today predict a MRF move tomorrow?"
4	"What will MRF's monthly price be 12 months from now?"

Code	Approach
(A)	VAR with Granger causality test
(B)	SARIMA on monthly prices
(C)	ARIMA on daily returns + directional forecast
(D)	GARCH conditional volatility forecast

Solution

① Correct Matches

Q	Answer	Reasoning
1	**(C)**	Directional prediction requires mean model — ARIMA sign = direction
2	**(D)**	Price swing magnitude = volatility — GARCH provides vol forecasts
3	**(A)**	Cross-asset predictability (BTC → MRF) = VAR Granger causality
4	**(B)**	Long-horizon monthly price with seasonal patterns = SARIMA

```
print("""
Model-Question Matching:

Q1 -> ARIMA (directional forecast):
  forecast = ARIMA(returns).forecast(1) -> sign ->
     direction
  Caveat: directional accuracy ~50% (efficient markets)

Q2 -> GARCH (volatility forecast):
  GARCH conditional vol x sqrt(22) = 1-month vol
  Answers "how much risk" not "which direction"

Q3 -> VAR + Granger causality:
  fit.test_causality(caused="MRF", causing=["BTC"])
  Significant p -> BTC past predicts MRF future

Q4 -> SARIMA (monthly price forecast):
  SARIMA(1,1,1)(1,1,0)[12].forecast(12)
  Handles non-stationarity (d=1) and seasonality (s=12)
""")
```

② Plain English

Model selection is the most important decision in time series analysis. Using GARCH to predict tomorrow's return direction is wrong — GARCH models variance, not mean. Using ARIMA to forecast month-ahead price is wrong — ARIMA on returns gives return forecasts, not price level forecasts with seasonal patterns. Each model class is designed for a specific type of question.

③ Common Mistake

Using ARIMA to answer Q2 (volatility question). ARIMA models the conditional mean of returns — its confidence intervals widen at longer horizons due to parameter uncertainty, but this is not the same as modelling time-varying conditional volatility. GARCH is the correct tool for answering "how large will price swings be?"

④ Real World Link

"Asking the right question of the right model" is a fundamental professional competency. A research analyst who uses GARCH to predict MRF's price direction has confused tools with questions. The mapping Q1→ARIMA direction, Q2→GARCH

vol, Q3→VAR causality, Q4→SARIMA price is the mental framework that professional quants use automatically.

Section D ★ Exam Style Projects — M5.024 to M5.027

M5.024 *Full Project — Bitcoin Complete Time Series Analysis* **EXAM**

Duration: 90 minutes | Marks: 40 | Dataset: BTC-USD

Part A — Data and Descriptive Statistics *[6 marks]*
Download BTC-USD (2019–2024). Compute: daily log returns (×100), annualised mean/std/Sharpe, skewness/kurtosis/Jarque-Bera test, maximum drawdown from price series, 95% HS VaR.

Part B — Stationarity and Model Identification *[8 marks]*
Run ADF + KPSS on price levels and log returns. Plot ACF and PACF (40 lags). Run Ljung-Box (20 lags) and ARCH LM test (10 lags) on returns. Create 2×2 diagnostic grid. Conclude: which models are appropriate?

Part C — ARIMA Mean Model *[8 marks]*
Use `auto_arima` to select order. Fit the model. Run full residual diagnostics. Forecast 30 days and plot with 90% and 95% confidence intervals.

Part D — GARCH Volatility Model *[10 marks]*
Fit GARCH(1,1) with t-distribution. Report all parameters. Compute and plot conditional volatility. Compute 99% VaR time series and backtest. Generate 22-day volatility forecast. Compute Expected Shortfall.

Part E — Investment Implications *[8 marks]*
Compute vol-targeted position size (target 30% annual vol). Compute 6-month forward P(BTC return > 0). Write a 10-line investment memo. Plot final 3-panel summary: price, conditional vol, VaR violations.

Solution

① Complete Solution

```
!pip install yfinance statsmodels arch pmdarima -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, scipy.stats as spstats
import warnings; warnings.filterwarnings('ignore')
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.graphics.tsaplots import plot_acf,
    plot_pacf
from statsmodels.stats.diagnostic import acorr_ljungbox
from arch.univariate import arch_lm_test
from arch import arch_model
from pmdarima import auto_arima
```

```
from statsmodels.tsa.arima.model import ARIMA

print("BTC-USD -- COMPLETE TIME SERIES ANALYSIS"); print("="*65)

# PART A
btc     = yf.download("BTC-USD", start="2019-01-01", end="2024-12-31", progress=False)
close   = btc["Close"].dropna()
returns = np.log(close).diff().dropna() * 100
n = len(returns); mu = returns.mean(); sigma = returns.std()
skew = spstats.skew(returns); kurt = spstats.kurtosis(returns)
jb_p = spstats.jarque_bera(returns).pvalue
roll_max = close.cummax(); max_dd = ((close-roll_max)/roll_max).min()
print(f"\nPART A | n={n} | Ann.Ret={mu*252:.1f}% | Ann.Vol={sigma*np.sqrt(252):.1f}%")
print(f"Sharpe={mu*252/(sigma*np.sqrt(252)):.4f} | Skew={skew:.4f} | ExKurt={kurt:.4f}")
print(f"JB p={jb_p:.6f} | MaxDD={max_dd*100:.1f}% | 95% HS VaR={np.percentile(returns,5):.4f}%")

# PART B
for s, lbl in [(close,"Price Level"),(returns,"Log Returns")]:
    adf = adfuller(s); kpss_r = kpss(s, regression="c", nlags="auto")
    krej = kpss_r[0] > kpss_r[3]["5%"]
    print(f"  {lbl}: ADF p={adf[1]:.6f} KPSS={'Non-stat' if krej else 'Stat'}")

fig, axes = plt.subplots(2,2, figsize=(16,10))
fig.suptitle("PART B -- BTC Identification Grid", fontsize=12)
returns.plot(ax=axes[0,0], color="steelblue", lw=0.5); axes[0,0].set_title("Log Returns")
plot_acf( returns, lags=40, ax=axes[0,1], zero=False); axes[0,1].set_title("ACF")
plot_pacf(returns, lags=40, ax=axes[1,0], zero=False); axes[1,0].set_title("PACF")
plot_acf(returns**2, lags=40, ax=axes[1,1], zero=False, color="orange")
axes[1,1].set_title("ACF Squared"); plt.tight_layout(); plt.show()

lb = acorr_ljungbox(returns, lags=20, return_df=True)
lm = arch_lm_test(returns, nlags=10)
```

```
print(f"LB min p={lb['lb_pvalue'].min():.4f} | ARCH LM p={lm.pvalue:.8f}")
print(f"ARIMA: {'Justified' if lb['lb_pvalue'].min()<0.05 else 'Limited'} | "
      f"GARCH: {'Required' if lm.pvalue<0.05 else 'Optional'}")

# PART C
auto_fit   = auto_arima(returns, start_p=0, max_p=2, start_q=0, max_q=2,
                        d=0, information_criterion="aic", stepwise=True,
                        suppress_warnings=True, error_action="ignore")
best_order = auto_fit.order
print(f"\nPART C | Selected: ARIMA{best_order} (AIC={auto_fit.aic():.2f})")
arima_fit = ARIMA(returns, order=best_order).fit()
lb_r  = acorr_ljungbox(arima_fit.resid.dropna(), lags=20, return_df=True)
lm_r  = arch_lm_test(arima_fit.resid.dropna(), nlags=10)
print(f"Residual LB p={lb_r['lb_pvalue'].min():.4f} | ARCH p={lm_r.pvalue:.4f}")

fc_obj = arima_fit.get_forecast(steps=30)
fc_mean= fc_obj.predicted_mean.values
fc_ci90= fc_obj.conf_int(alpha=0.10); fc_ci95 = fc_obj.conf_int(alpha=0.05)

fig, ax = plt.subplots(figsize=(14,5))
ax.plot(range(60), returns[-60:].values, color="steelblue", lw=0.8)
ax.plot(range(60,90), fc_mean, "r-", lw=2, label="30d Forecast")
ax.fill_between(range(60,90), fc_ci95.iloc[:,0], fc_ci95.iloc[:,1],
                color="red", alpha=0.15, label="95% CI")
ax.fill_between(range(60,90), fc_ci90.iloc[:,0], fc_ci90.iloc[:,1],
                color="red", alpha=0.25, label="90% CI")
ax.axhline(0, color="black", lw=0.8, linestyle="--")
ax.axvline(60, color="gray", lw=1.5, linestyle=":")
ax.set_title(f"PART C -- ARIMA{best_order}: 30-Day Return Forecast")
ax.legend(fontsize=9); plt.tight_layout(); plt.show()

# PART D
garch_fit = arch_model(returns, mean="Constant", vol="GARCH",
```

```
                          p=1, q=1, dist="t").fit(disp="off")
omega=garch_fit.params["omega"]; alpha_g=garch_fit.params["alpha[1]"]
beta_g=garch_fit.params["beta[1]"]; nu=garch_fit.params["nu"]
perst=alpha_g+beta_g; hl=np.log(0.5)/np.log(perst)
lrv=omega/(1-perst); lrvol=np.sqrt(lrv)
latest_vol=garch_fit.conditional_volatility.iloc[-1]
cond_v = garch_fit.conditional_volatility; mu_g = garch_fit.params["mu"]

print(f"\nPART D | alpha={alpha_g:.4f} beta={beta_g:.4f} nu={nu:.2f}")
print(f"Half-life={hl:.1f}d | LR vol={lrvol*np.sqrt(252):.2f}% | "
      f"Current vol={latest_vol*np.sqrt(252):.2f}%")

var99 = mu_g + spstats.norm.ppf(0.01)*cond_v
viols = returns < var99
print(f"99% VaR backtest: {viols.sum()} violations | rate={viols.mean()*100:.2f}%")

fc_g  = garch_fit.forecast(horizon=22, reindex=False)
vol22 = np.sqrt(fc_g.variance.iloc[0].values)
t_alpha = spstats.t.ppf(0.01, df=nu); pdf_val = spstats.t.pdf(t_alpha, df=nu)
es99 = mu_g + (-(pdf_val/0.01)*(nu+t_alpha**2)/(nu-1)) * cond_v.iloc[-1]
print(f"22-day avg vol forecast: {vol22.mean():.4f}%/day | 99% ES={es99:.4f}%")

# PART E
target_vol = 30.0; curr_vol_a = latest_vol*np.sqrt(252)
position   = min(target_vol/curr_vol_a, 2.0)
h = 126; cum_fc = np.sum(fc_mean[:30]); cum_vol = vol22.mean()*np.sqrt(h)
prob_pos = 1 - spstats.norm.cdf(0, loc=cum_fc, scale=cum_vol)

print(f"\nPART E | Target vol=30% | Current vol={curr_vol_a:.1f}%")
print(f"Vol-targeted position: {position:.4f}x ({position*100:.1f}% of capital)")
print(f"6-month P(return>0): {prob_pos*100:.0f}%")

fig, axes = plt.subplots(3,1, figsize=(18,16))
fig.suptitle("BTC-USD -- Research Summary Dashboard", fontsize=14)
```

```
close.plot(ax=axes[0], color="orange", lw=1); axes[0].set_title("BTC-USD Price (USD)")
cond_v.plot(ax=axes[1], color="red", lw=0.8)
axes[1].axhline(lrvol, color="black", lw=1.5, linestyle="--",
                label=f"LR vol={lrvol:.2f}%")
axes[1].set_title("GARCH Conditional Volatility"); axes[1].legend()
axes[2].plot(returns, color="steelblue", lw=0.5, alpha=0.7, label="Returns")
axes[2].plot(var99,   color="red", lw=1, label="99% VaR")
axes[2].scatter(returns[viols].index, returns[viols],
                color="darkred", s=10, zorder=5,
                   label="Violation")
axes[2].set_title("Returns vs 99% VaR with Violations"); axes[2].legend(fontsize=8)
plt.tight_layout(); plt.show()

print(f"""

  INVESTMENT MEMO: Bitcoin (BTC-USD)

  1. BTC returns are near-white-noise in mean (ARIMA{best_order}),
     but display extreme volatility clustering (GARCH essential).
  2. t-distribution required: nu={nu:.1f} confirms fat tails.
  3. Current vol regime: {curr_vol_a:.0f}% annual
     {'ABOVE' if curr_vol_a>lrvol*np.sqrt(252) else 'BELOW'} long-run avg ({lrvol*np.sqrt(252):.0f}%).
  4. Vol-targeted position: {position:.2f}x (target 30% portfolio vol).
  5. 99% VaR violation rate: {viols.mean()*100:.2f}% (Basel target: <1.6%).
  6. ES at 99%: {es99:.2f}%  (expected loss on worst 1% of days).
  7. Shock half-life: {hl:.0f} trading days (vol shocks persist for weeks).
  8. 6-month P(positive return): {prob_pos*100:.0f}% (modest positive drift).
  9. Key risk: structural breaks (regulatory, exchange failures).
 10. Recommendation: Allocate {position*100:.0f}% of vol budget to BTC;
     use GARCH-VaR for daily risk monitoring.

""")
```

④ Real World Link

This research note structure is the standard template for quantitative equity/crypto research. Each part answers one fundamental question. Part E's investment memo is the deliverable that reaches portfolio managers — it synthesises statistical analysis into actionable intelligence. The ability to write this memo clearly with specific numbers and honest caveats is the defining skill of a senior data scientist in finance.

M5.025 *Full Project — MRF Tyres Investment-Grade Analysis* EXAM

Duration: 90 minutes | Marks: 40 | Dataset: MRF.NS

Part A (6 marks): Download MRF.NS (2019–2024), daily and monthly. Compute descriptive statistics. Compare MRF's volatility profile to BTC in a joint table. Test normality. Classify MRF as low/medium/high volatility asset relative to the three course assets.

Part B (8 marks): Decompose monthly MRF prices (multiplicative, period=12). Plot all four components. Identify: (a) secular trend character, (b) peak seasonal months, (c) residual structure.

Part C (8 marks): Complete Box-Jenkins: ADF, ACF/PACF, auto-order selection, fit, residual diagnostics, 30-day forward return forecast.

Part D (10 marks): Fit GARCH(1,1) with t-distribution. Compare to GJR-GARCH. Select better model by AIC. Full parameter table. Compute regime classifications (low/medium/high vol). 22-day vol forecast.

Part E (8 marks): Using SARIMA(1,1,1)(1,1,0)[12] on monthly prices: generate 12-month price forecast. Use GARCH-Monte Carlo to simulate 1000 price paths. Compute Bear/Base/Bull cases. Plot fan chart. Write a 5-line price target statement.

Solution

① Complete Solution

```
!pip install yfinance statsmodels arch pmdarima -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, scipy.stats as spstats
import warnings; warnings.filterwarnings('ignore')
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf,
   plot_pacf
from statsmodels.stats.diagnostic import acorr_ljungbox
from arch.univariate import arch_lm_test
from arch import arch_model
from pmdarima import auto_arima
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.statespace.sarimax import SARIMAX
```

```

print("MRF TYRES -- INVESTMENT-GRADE QUANTITATIVE ANALYSIS"); print("="*65)

mrf      = yf.download("MRF.NS",  start="2019-01-01", end="2024-12-31", progress=False)
btc      = yf.download("BTC-USD", start="2019-01-01", end="2024-12-31", progress=False)
gold     = yf.download("GC=F",    start="2019-01-01", end="2024-12-31", progress=False)

mrf_close  = mrf["Close"].dropna()
mrf_ret    = np.log(mrf_close).diff().dropna() * 100
mrf_monthly= mrf_close.resample("ME").last()

# PART A
def stats(ret, name):
    return {"Asset":name,
            "Ann. Return (%)": round(ret.mean()*252,2),
            "Ann. Vol (%)":
               round(ret.std()*np.sqrt(252),2),
            "Sharpe":
               round((ret.mean()*252)/(ret.std()*np.sqrt(252)),4),
            "Skewness":        round(spstats.skew(ret),3),
            "Ex. Kurt":
               round(spstats.kurtosis(ret),3),
            "JB p-value":
               round(spstats.jarque_bera(ret).pvalue,6)}

rows = [stats(mrf_ret, "MRF.NS"),
        stats(np.log(btc["Close"]).diff().dropna()*100,
           "BTC-USD"),
        stats(np.log(gold["Close"]).diff().dropna()*100,
           "GC=F")]
stats_df = pd.DataFrame(rows).set_index("Asset")
print("\nPART A"); print(stats_df.to_string())
mrf_vol = stats_df.loc["MRF.NS","Ann. Vol (%)"]
all_vols = {"BTC":stats_df.loc["BTC-USD","Ann. Vol (%)"],
            "MRF":mrf_vol,
               "Gold":stats_df.loc["GC=F","Ann. Vol (%)"]}
rank = sorted(all_vols.values()).index(mrf_vol) + 1
print(f"\nMRF volatility rank: {rank}/3  Classification: "
      f"{'Low' if rank==1 else ('Medium' if rank==2 else
         'High')} ({mrf_vol:.1f}% p.a.)")

# PART B
result = seasonal_decompose(mrf_monthly, model="multiplicative", period=12)
result.plot()
```

```
plt.suptitle("PART B -- MRF Monthly Multiplicative Decomposition", fontsize=11)
plt.tight_layout(); plt.show()
trend = result.trend.dropna()
cagr  = (trend.iloc[-1]/trend.iloc[0])**((trend.index[-1]-trend.index[0]).days/365.25) - 1
print(f"\nPART B | Trend CAGR: {cagr*12*100:.1f}% per year")
seas_f = result.seasonal
first_yr = seas_f[seas_f.index.year == seas_f.index[12].year]
months = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"]
print("  Seasonal factors:")
for mname, (_, val) in zip(months, first_yr.items()):
    print(f"    {mname}: {val:.4f}  {'+' if val>1 else '-'}")

# PART C
adf_mrf = adfuller(mrf_ret)
print(f"\nPART C | ADF p={adf_mrf[1]:.6f} -> {'Stationary' if adf_mrf[1]<0.05 else 'Non-stationary'}")
auto_mrf  = auto_arima(mrf_ret, start_p=0, max_p=2, start_q=0, max_q=2,
                       d=0, information_criterion="aic", stepwise=True,
                       suppress_warnings=True, error_action="ignore")
mrf_order = auto_mrf.order
print(f"Selected: ARIMA{mrf_order} (AIC={auto_mrf.aic():.2f})")
arima_mrf = ARIMA(mrf_ret, order=mrf_order).fit()
lb_m = acorr_ljungbox(arima_mrf.resid.dropna(), lags=20, return_df=True)
lm_m = arch_lm_test(arima_mrf.resid.dropna(), nlags=10)
print(f"Residual LB p={lb_m['lb_pvalue'].min():.4f} | ARCH p={lm_m.pvalue:.4f}")
fc_30_m = arima_mrf.get_forecast(steps=30)
print(f"30-day cum. forecast: {fc_30_m.predicted_mean.sum():+.4f}%")

# PART D
garch_mrf = arch_model(mrf_ret, mean="Constant", vol="GARCH", p=1, q=1, dist="t").fit(disp="off")
gjr_mrf   = arch_model(mrf_ret, mean="Constant", vol="GARCH", p=1, o=1, q=1, dist="t").fit(disp="off")
```

```
gamma_g   = gjr_mrf.params["gamma[1]"]
print(f"\nPART D | GARCH AIC={garch_mrf.aic:.2f} | GJR AIC={gjr_mrf.aic:.2f}")
print(f"GJR gamma={gamma_g:.4f} "
      f"({'Classical leverage' if gamma_g>0.01 else ('Inverse leverage' if gamma_g<-0.01 else 'Symmetric')})")
best_g = garch_mrf if garch_mrf.aic < gjr_mrf.aic else gjr_mrf
cond_v_m = best_g.conditional_volatility
q25, q75 = cond_v_m.quantile(0.25), cond_v_m.quantile(0.75)
regime = pd.cut(cond_v_m, bins=[0,q25,q75,np.inf], labels=["Low","Medium","High"])
print("Regime distribution:"); print(regime.value_counts().sort_index())
fc_gm  = best_g.forecast(horizon=22, reindex=False)
vol22m = np.sqrt(fc_gm.variance.iloc[0].values)
print(f"22-day vol forecast: {vol22m.mean():.4f}%/day ({vol22m.mean()*np.sqrt(252):.2f}%/yr)")

# PART E
sarima_fit = SARIMAX(mrf_monthly, order=(1,1,1), seasonal_order=(1,1,0,12),
                     enforce_stationarity=False, enforce_invertibility=False).fit(disp=False)
sarima_fc  = sarima_fit.forecast(steps=12)

np.random.seed(42); n_sims=1000; n_months=12
last_p = mrf_monthly.iloc[-1]
avg_mvol = vol22m.mean()*np.sqrt(21)
mu_monthly = mrf_ret.mean()*21/100
sim_paths = np.zeros((n_months, n_sims))
for s in range(n_sims):
    price = last_p
    for m in range(n_months):
        price = price*(1 + mu_monthly + avg_mvol*np.random.randn()/100)
        sim_paths[m,s] = price

bear = np.percentile(sim_paths,10,axis=1)
base = np.percentile(sim_paths,50,axis=1)
bull = np.percentile(sim_paths,90,axis=1)

fig, ax = plt.subplots(figsize=(16,7))
mrf_monthly[-24:].plot(ax=ax, color="steelblue", lw=2, label="Historical MRF")
sarima_fc.plot(ax=ax, color="red", lw=2, label="SARIMA Forecast")
```

```
month_idx = pd.date_range(mrf_monthly.index[-1], periods=13, freq="ME")[1:]
ax.fill_between(month_idx, bear, bull,  color="orange", alpha=0.3, label="MC 80% range")
ax.fill_between(month_idx, np.percentile(sim_paths,25,axis=1), np.percentile(sim_paths,75,axis=1), color="orange", alpha=0.5, label="MC 50% range")
ax.plot(month_idx, base, "ko-", markersize=4, lw=1.5, label="MC Median")
ax.plot(month_idx, bear, "r--", lw=1, label=f"Bear (Rs.{bear[-1]:,.0f})")
ax.plot(month_idx, bull, "g--", lw=1, label=f"Bull (Rs.{bull[-1]:,.0f})")
ax.axvline(mrf_monthly.index[-1], color="gray", lw=1.5, linestyle=":")
ax.set_title("PART E -- MRF Tyres: 12-Month Fan Chart (SARIMA + GARCH Monte Carlo)")
ax.set_ylabel("Rs. INR"); ax.legend(fontsize=8); plt.tight_layout(); plt.show()

print(f"""

  MRF TYRES: 12-MONTH PRICE TARGET STATEMENT

  SARIMA point estimate: Rs.{sarima_fc.iloc[-1]:,.0f}
    ({(sarima_fc.iloc[-1]/last_p-1)*100:+.1f}% vs current Rs.{last_p:,.0f})
  Bear case (10th pctile): Rs.{bear[-1]:,.0f}
    ({(bear[-1]/last_p-1)*100:+.1f}%)
  Bull case (90th pctile): Rs.{bull[-1]:,.0f}
    ({(bull[-1]/last_p-1)*100:+.1f}%)
  Key risk: Structural break in tyre demand or rubber price shock.

""")
```

M5.026 *Full Project — Gold Futures Macro Risk System* EXAM

Duration: 90 minutes | Marks: 40 | Dataset: GC=F

Part A (8 marks): Full EDA on Gold daily and monthly: descriptive stats, stationarity tests, seasonal decomposition with seasonal factor table, Ljung-Box and ARCH LM on daily returns.

Part B (8 marks): SARIMA model selection grid ($p, q \in \{0, 1, 2\}$, seasonal $P, Q \in \{0, 1\}$) on monthly Gold prices. AIC comparison table. Best model fit and residual validation.

Part C (8 marks): GARCH(1,1) with GJR extension on daily returns. Compare symmetric vs asymmetric specification. Compute time-varying 99% VaR. Backtest and apply Kupiec POF test.

Part D (8 marks): Combine SARIMA (monthly trend/seasonal forecast) with GARCH (daily volatility forecast) to build a multi-scale risk-return framework. Generate: (1) 12-month price path from SARIMA, (2) daily VaR around that path from GARCH, (3) fan chart combining both.

Part E (8 marks): Write a complete Gold investment memo: seasonal timing, volatility regime, 12-month price target range, key risk factors, recommended hedge ratio for a 20% Gold allocation.

Solution

① Complete Solution

```
!pip install yfinance statsmodels arch -q
import yfinance as yf, numpy as np, pandas as pd
import matplotlib.pyplot as plt, scipy.stats as spstats
import itertools, warnings;
   warnings.filterwarnings('ignore')
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.stats.diagnostic import acorr_ljungbox
from arch.univariate import arch_lm_test
from arch import arch_model
from statsmodels.tsa.statespace.sarimax import SARIMAX

print("GOLD FUTURES -- MACRO RISK AND FORECASTING
   SYSTEM"); print("="*65)

gold    = yf.download("GC=F", start="2019-01-01",
   end="2024-12-31", progress=False)
close   = gold["Close"].dropna(); returns =
   np.log(close).diff().dropna()*100
monthly = close.resample("ME").last()

# PART A
n=len(returns); mu=returns.mean(); sigma=returns.std()
skew=spstats.skew(returns); kurt=spstats.kurtosis(returns)
jb_p=spstats.jarque_bera(returns).pvalue
print(f"\nPART A | n={n} | Ann.Ret={mu*252:.2f}% |
   Ann.Vol={sigma*np.sqrt(252):.2f}%")
print(f"Skew={skew:.4f} | ExKurt={kurt:.4f} | JB
   p={jb_p:.6f}")

for s, lbl in [(close,"Price"),(returns,"Returns")]:
    adf=adfuller(s);
       kpss_r=kpss(s,regression="c",nlags="auto")
```

```
    krej=kpss_r[0]>kpss_r[3]["5%"]
    print(f"  {lbl}: ADF p={adf[1]:.4f} KPSS={'Non-stat' if krej else 'Stat'}")

decomp = seasonal_decompose(monthly, model="multiplicative", period=12)
result = decomp.seasonal; first_yr = result[result.index.year==result.index[12].year]
months = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","D
print("\n  Gold Seasonal Factors:")
for mname, (_, val) in zip(months, first_yr.items()):
    bar = "#"*int(abs(val-1)*200)
    print(f"    {mname}: {val:.4f}  {'+' if val>1 else '-'}{bar}")

lb_g=acorr_ljungbox(returns,lags=20,return_df=True); lm_g=arch_lm_test(returns,nlags=10)
print(f"LB min p={lb_g['lb_pvalue'].min():.4f} | ARCH LM p={lm_g.pvalue:.8f}")

# PART B
train_m=monthly[:-12]; test_m=monthly[-12:]
aic_results=[]
for p,q,P,Q in itertools.product([0,1,2],[0,1,2],[0,1],[0,1]):
    try:
        f = SARIMAX(train_m, order=(p,1,q), seasonal_order=(P,1,Q,12),
                    enforce_stationarity=False, enforce_invertibility=False).fit(disp=False)
        aic_results.append({"Order":f"({p},1,{q})({P},1,{Q})[12]",
                            "p":p,"q":q,"P":P,"Q":Q,"AIC":round(f.aic,2
    except Exception: pass

aic_df  = pd.DataFrame(aic_results).sort_values("AIC").reset_index(drop=True)
print("\nPART B -- Top 5 SARIMA models:")
print(aic_df[["Order","AIC","BIC"]].head(5).to_string(index=False))

best_r = aic_df.iloc[0]
best_s = SARIMAX(train_m, order=(int(best_r["p"]),1,int(best_r["q"])),
                 seasonal_order=(int(best_r["P"]),1,int(best_r["Q"]),12
                 enforce_stationarity=False,enforce_invertibility=False
lb_s  = acorr_ljungbox(best_s.resid.dropna(), lags=15, return_df=True)
rmse_s= np.sqrt(np.mean((test_m.values - best_s.forecast(12).values)**2))
```

```
print(f"Best: {best_r['Order']} | Residual LB p={lb_s['lb_pvalue'].min():.4f} | Test RMSE=${rmse_s:,.0f}")

# PART C
garch_sym = arch_model(returns, mean="Constant", vol="GARCH", p=1, q=1, dist="t").fit(disp="off")
garch_gjr = arch_model(returns, mean="Constant", vol="GARCH", p=1, o=1, q=1, dist="t").fit(disp="off")
gamma_gold= garch_gjr.params["gamma[1]"]
print(f"\nPART C | GARCH AIC={garch_sym.aic:.2f} | GJR AIC={garch_gjr.aic:.2f}")
print(f"GJR gamma={gamma_gold:.4f} "
      f"({'Classical leverage' if gamma_gold>0.01 else ('Inverse' if gamma_gold<-0.01 else 'Symmetric')})")
best_gg   = garch_sym if garch_sym.aic < garch_gjr.aic else garch_gjr
nu_gold   = best_gg.params["nu"]; mu_gold = best_gg.params["mu"]
cond_v_g  = best_gg.conditional_volatility
var99_g   = mu_gold + spstats.t.ppf(0.01, df=nu_gold)*cond_v_g
viols_g   = returns < var99_g; viol_rt_g = viols_g.mean()
T=len(returns); x=viols_g.sum(); ph=x/T
if ph>0 and ph<1:
    lr_k = -2*(x*np.log(0.01)+(T-x)*np.log(0.99)-x*np.log(ph)-(T-x)*np.log(1
    p_k  = 1-spstats.chi2.cdf(lr_k, df=1)
else: p_k = np.nan
print(f"99% VaR rate={viol_rt_g*100:.2f}% | Kupiec p={p_k:.4f} "
      f"-> {'OK' if p_k>0.05 else 'Rejected'}")

# PART D
sarima_full = SARIMAX(monthly, order=(int(best_r["p"]),1,int(best_r["q"])),
                      seasonal_order=(int(best_r["P"]),1,int(best_r["Q"
                      enforce_stationarity=False,enforce_invertibility=
sarima_fc_g = sarima_full.forecast(steps=12)
fc_vol_g    = best_gg.forecast(horizon=22, reindex=False)
avg_vol_g   = np.sqrt(fc_vol_g.variance.iloc[0].values).mean()

np.random.seed(123); n_sims=1000; n_days=252
last_pg = close.iloc[-1]; daily_paths = np.zeros((n_days, n_sims))
for s in range(n_sims):
    price = last_pg
```

```
    for d in range(n_days):
        price = price*(1 + mu_gold/100 +
           avg_vol_g/100*np.random.randn())
        daily_paths[d,s] = price

mc_monthly = np.zeros((12, n_sims))
for m in range(12):
    day_idx = min(int((m+1)*21), n_days-1)
    mc_monthly[m] = daily_paths[day_idx]

bear_g = np.percentile(mc_monthly,10,axis=1)
base_g = np.percentile(mc_monthly,50,axis=1)
bull_g = np.percentile(mc_monthly,90,axis=1)

fig, ax = plt.subplots(figsize=(16,7))
monthly[-24:].plot(ax=ax, color="gold", lw=2,
   label="Historical Gold")
sarima_fc_g.plot(ax=ax, color="red", lw=2, label="SARIMA
   Forecast")
ni = pd.date_range(monthly.index[-1], periods=13,
   freq="ME")[1:]
ax.fill_between(ni, bear_g, bull_g, color="gold",
   alpha=0.25, label="MC 80% range")
ax.fill_between(ni, np.percentile(mc_monthly,25,axis=1),
                np.percentile(mc_monthly,75,axis=1),
                   color="gold", alpha=0.45, label="MC 50%
                   range")
ax.plot(ni, base_g, "ko-", markersize=4, lw=1.5, label="MC
   Median")
ax.plot(ni, bear_g, "r--", lw=1, label=f"Bear
   (${bear_g[-1]:,.0f})")
ax.plot(ni, bull_g, "g--", lw=1, label=f"Bull
   (${bull_g[-1]:,.0f})")
ax.axvline(monthly.index[-1], color="gray", lw=1.5,
   linestyle=":")
ax.set_title("PART D -- Gold: SARIMA + GARCH Monte Carlo
   Fan Chart")
ax.set_ylabel("USD/oz"); ax.legend(fontsize=8);
   plt.tight_layout(); plt.show()

# PART E
perst_g = best_gg.params["alpha[1]"] +
   best_gg.params["beta[1]"]
hl_g    = np.log(0.5)/np.log(perst_g)
curr_vg = cond_v_g.iloc[-1]*np.sqrt(252)
lrv_g   = best_gg.params["omega"]/(1-perst_g)
lrvol_g = np.sqrt(lrv_g)*np.sqrt(252)
peak_m  = months[first_yr.values.argmax()]

```

```
print(f"""
        PART E: GOLD INVESTMENT MEMO

  SEASONAL TIMING: Peak demand month = {peak_m}
  (factor={first_yr.values.max():.4f},
     {(first_yr.values.max()-1)*100:.1f}% above annual avg)
  Strategy: Scale Gold exposure 1-2 months before seasonal
     peak.

  VOLATILITY REGIME: Current annual vol = {curr_vg:.1f}%
     (LR avg: {lrvol_g:.1f}%)
  Regime: {'HIGH -- reduce position' if
     curr_vg>lrvol_g*1.2 else ('LOW -- potential to add'
     if curr_vg<lrvol_g*0.8 else 'NORMAL -- maintain
     allocation')}
  Shock half-life: {hl_g:.0f} trading days

  12-MONTH PRICE TARGET:
  Bear (10th pctile): ${bear_g[-1]:,.0f}
     ({(bear_g[-1]/close.iloc[-1]-1)*100:+.1f}%)
  Base (MC median)  : ${base_g[-1]:,.0f}
     ({(base_g[-1]/close.iloc[-1]-1)*100:+.1f}%)
  SARIMA point est  : ${sarima_fc_g.iloc[-1]:,.0f}
     ({(sarima_fc_g.iloc[-1]/close.iloc[-1]-1)*100:+.1f}%)
  Bull (90th pctile): ${bull_g[-1]:,.0f}
     ({(bull_g[-1]/close.iloc[-1]-1)*100:+.1f}%)

  KEY RISK FACTORS:
  1. Federal Reserve rate decisions (Gold sensitive to
     real rates)
  2. USD strength (inverse relationship with Gold)
  3. Geopolitical escalation (demand spike risk)
  4. Central bank buying (PBOC and RBI flows)
  5. Structural break: de-correlation from safe-haven role

  HEDGE RATIO (20% Gold portfolio):
  Vol-targeted weight at 10% port. vol:
     {min(10/curr_vg,1)*100:.1f}%
  (Reduce from 20% if current vol regime is HIGH)
""")
```

M5.027 *Full Project — Three-Asset System: Complete Portfolio Analytics* **EXAM**

Duration: 2 hours | Marks: 50 | Dataset: BTC-USD, MRF.NS, GC=F

Part A (8 marks): Download all three assets (2019–2024). Build a unified 4×3 chart matrix: rows = (price, log returns, squared returns, rolling 90-day vol), columns = (BTC, MRF, Gold). Print a cross-asset statistical comparison table.
Part B (8 marks): ADF + KPSS on returns (table). Fit 3-variable VAR. Com-

plete Granger causality matrix. FEVD at horizon 10. Write a cross-asset dynamics summary.

Part C (10 marks): For each asset, fit the appropriate model. BTC: GARCH(1,1) with t-distribution. MRF: ARIMA + GARCH two-stage. Gold: SARIMA on monthly + GARCH on daily. Report AIC, parameters, residual validation, 22-day forecast.

Part D (12 marks): Using GARCH conditional volatilities and DCC rolling correlation: compute minimum-variance and inverse-vol weights monthly. Plot both weight time series. Compute portfolio statistics. Compare to 1/3-1/3-1/3 equal-weight.

Part E (12 marks): Portfolio 99% VaR and ES (GARCH). Simulate 1000 portfolio paths (6 months). Fan chart. Three stress scenarios. Write a 15-line institutional investment committee summary.

Solution

① Complete Solution

```
!pip install yfinance statsmodels arch pmdarima scipy -q
import yfinance as yf, numpy as np, pandas as pd,
   scipy.stats as spstats
import matplotlib.pyplot as plt
from scipy.optimize import minimize
import warnings; warnings.filterwarnings('ignore')
from statsmodels.tsa.stattools import adfuller, kpss
from statsmodels.tsa.vector_ar.var_model import VAR
from arch import arch_model, arch_lm_test
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.statespace.sarimax import SARIMAX

print("THREE-ASSET PORTFOLIO CAPSTONE: BTC * MRF * GOLD");
   print("="*65)

# Data
tickers = {"BTC":"BTC-USD","MRF":"MRF.NS","Gold":"GC=F"}
closes  = {}; rets = {}
for name, ticker in tickers.items():
    raw = yf.download(ticker, start="2019-01-01",
       end="2024-12-31", progress=False)
    closes[name] = raw["Close"].dropna()
    rets[name]   = np.log(closes[name]).diff().dropna() *
       100

ret_df = pd.DataFrame(rets).dropna(); assets =
   list(tickers.keys())

# PART A
fig, axes = plt.subplots(4, 3, figsize=(21, 20))
```

```
fig.suptitle("Three-Asset Dashboard: BTC * MRF * Gold", fontsize=14, fontweight="bold")
colors = {"BTC":"#F7931A","MRF":"#1E90FF","Gold":"#FFD700"}

for col, asset in enumerate(assets):
    close_a = closes[asset]; ret_a = rets[asset]; c = colors[asset]
    axes[0,col].plot(close_a, color=c, lw=0.8); axes[0,col].set_title(f"{asset} Price")
    axes[1,col].plot(ret_a, color=c, lw=0.4, alpha=0.7)
    axes[1,col].axhline(0, color="black", lw=0.6); axes[1,col].set_title(f"{asset} Returns")
    axes[2,col].fill_between(ret_a.index, ret_a**2, 0, color=c, alpha=0.5)
    axes[2,col].set_title(f"{asset} Squared Returns (ARCH)")
    rv = ret_a.rolling(90).std()*np.sqrt(252)
    axes[3,col].plot(rv, color=c, lw=1)
    axes[3,col].axhline(rv.mean(), color="black", lw=1, linestyle="--")
    axes[3,col].set_title(f"{asset} Rolling 90d Annual Vol")
plt.tight_layout(); plt.show()

stat_rows = []
for asset, ret_a in rets.items():
    stat_rows.append({"Asset":asset,
        "Ann Ret":round(ret_a.mean()*252,2),
        "Ann Vol":round(ret_a.std()*np.sqrt(252),2),
        "Sharpe":
           round((ret_a.mean()*252)/(ret_a.std()*np.sqrt(252)),3),
        "Skew":   round(spstats.skew(ret_a),3),
        "ExKurt": round(spstats.kurtosis(ret_a),3),
        "MaxLoss":round(ret_a.min(),3)})
print("\nPART A -- Cross-Asset Statistics:")
print(pd.DataFrame(stat_rows).set_index("Asset").to_string())

# PART B
print("\nPART B -- Stationarity Summary:")
print(f"  {'Asset':8} {'ADF p':>10} {'Stationary?':>14}")
for asset, ret_a in rets.items():
    adf = adfuller(ret_a)
    print(f"  {asset:8} {adf[1]:>10.6f} {'YES' if adf[1]<0.05 else 'NO':>14}")

var_fit = VAR(ret_df).fit(maxlags=5, ic="aic")
print(f"\n  VAR({var_fit.k_ar}) fitted | AIC={var_fit.aic:.2f}")

```

```
print(f"\n  Granger Causality Matrix (p-values):")
print(f"  {'':8}", end="")
for a in assets: print(f"{a:>10}", end="")
print()
for caused in assets:
    print(f"  {caused:8}", end="")
    for causing in assets:
        if caused == causing: print(f"{'---':>10}", end="")
        else:
            gc = var_fit.test_causality(caused=caused,
                causing=[causing])
            m  = "*" if gc.pvalue < 0.05 else " "
            print(f"{gc.pvalue:>9.4f}{m}", end="")
    print()
print("  (* = significant at 5%)")

fevd = var_fit.fevd(10)
fevd_10 = pd.DataFrame(fevd.decomp[9], index=assets,
   columns=assets)
print(f"\n  FEVD at Horizon 10:");
   print(fevd_10.round(4).to_string())
most_exog = fevd_10.values.diagonal().argmax()
print(f"  Most exogenous: {assets[most_exog]}
   ({fevd_10.values[most_exog,most_exog]*100:.1f}%
   own-shock)")

print(f"""
  Cross-Asset Dynamics Summary:
  VAR({var_fit.k_ar}) reveals limited daily Granger
     causality across assets.
  FEVD shows each asset primarily self-driven (own-shock
     typically >90%).
  This confirms genuine diversification: assets are
     largely independent.
  For meaningful cross-asset dynamics, weekly/monthly VAR
     recommended.
""")

# PART C
garch_fits = {}; model_results = {}
for asset in assets:
    ret_a = rets[asset]; print(f"\n  {asset}:")
    fit = arch_model(ret_a, mean="Constant", vol="GARCH",
                     p=1, q=1, dist="t").fit(disp="off")
    garch_fits[asset] = fit
    a = fit.params["alpha[1]"]; b = fit.params["beta[1]"]
    model_results[asset] = {
        "model":"GARCH(1,1)-t","AIC":fit.aic,
        "persistence":a+b,
```

```
        "current_vol_ann":fit.conditional_volatility.iloc[-1]*np.sqrt(25
        "fit":fit}
    print(f"    GARCH(1,1)-t | AIC={fit.aic:.2f} |
       alpha+beta={a+b:.4f}")
    print(f"    Current annual vol:
       {fit.conditional_volatility.iloc[-1]*np.sqrt(252):.2f}%")

print("\n  Model Summary:")
print(f"  {'Asset':8} {'Model':25} {'a+b':>8} {'Curr
   Vol':>12}")
for asset, res in model_results.items():
    print(f"  {asset:8} {res['model']:25}
       {res['persistence']:>8.4f} "
          f"{res['current_vol_ann']:>11.2f}%")

# PART D
monthly_idx = ret_df.resample("ME").last().index
lam = 0.94
z_arr = {a: garch_fits[a].std_resid.dropna() for a in
   assets}
z_df  = pd.DataFrame(z_arr).dropna()
n_z   = len(z_df); Q_t = np.cov(z_df.values.T); roll_corrs
   = []

for t in range(1, n_z):
    z  = z_df.iloc[t-1].values.reshape(-1,1)
    Q_t = lam*Q_t + (1-lam)*(z@z.T)
    D_inv = np.diag(1/np.sqrt(np.diag(Q_t)))
    roll_corrs.append(D_inv@Q_t@D_inv)

def port_weights_mv(vols, corr_mat):
    n   = len(vols); cov =
       np.diag(vols)@corr_mat@np.diag(vols)
    r   = minimize(lambda w: np.sqrt(w@cov@w),
       np.ones(n)/n,
                   bounds=[(0,1)]*n,
                   constraints=[{"type":"eq","fun":lambda
                      w: w.sum()-1}], method="SLSQP")
    return r.x if r.success else np.ones(n)/n

mv_weights = []; ms_weights = []; port_dates = []
for dt in monthly_idx:
    if dt > ret_df.index[252]:
        vols = np.array([
            garch_fits[a].conditional_volatility.reindex([dt],
               method="nearest")[0]
            if dt in
               garch_fits[a].conditional_volatility.index
```

```
            else
                garch_fits[a].conditional_volatility.mean()
            for a in assets])
        rho = roll_corrs[-1] if roll_corrs else np.eye(3)
        w_mv = port_weights_mv(vols, rho)
        inv_vol = 1/vols; w_ms = inv_vol/inv_vol.sum()
        mv_weights.append(w_mv); ms_weights.append(w_ms);
            port_dates.append(dt)

mv_df = pd.DataFrame(mv_weights, index=port_dates,
    columns=assets)
ms_df = pd.DataFrame(ms_weights, index=port_dates,
    columns=assets)

fig, axes = plt.subplots(2,1, figsize=(16,10))
mv_df.plot.bar(ax=axes[0], stacked=True,
    colormap="viridis", width=0.8, legend=True)
axes[0].set_title("Minimum Variance Weights (Monthly)")
axes[0].set_xticklabels([d.strftime("%b%Y") for d in
    mv_df.index], rotation=45, fontsize=7)
ms_df.plot.bar(ax=axes[1], stacked=True,
    colormap="plasma", width=0.8, legend=True)
axes[1].set_title("Inverse-Vol Weights (Monthly)")
axes[1].set_xticklabels([d.strftime("%b%Y") for d in
    ms_df.index], rotation=45, fontsize=7)
plt.tight_layout(); plt.show()

def portfolio_returns(weights_df, returns_df):
    port_ret = []
    for i, dt in enumerate(weights_df.index[:-1]):
        next_dt = weights_df.index[i+1]
        w = weights_df.iloc[i].values
        month_r = returns_df.loc[dt:next_dt].mean().values
        port_ret.append(w @ month_r)
    return pd.Series(port_ret)

port_mv = portfolio_returns(mv_df, ret_df)
port_ms = portfolio_returns(ms_df, ret_df)

def perf(ret_s, name):
    cum = (1+ret_s/100).prod()-1
    ann = ret_s.mean()*252; vol = ret_s.std()*np.sqrt(252)
    sh  = ann/vol if vol>0 else 0
    peak = (1+ret_s/100).cumprod().cummax()
    dd   = ((1+ret_s/100).cumprod()/peak-1).min()
    print(f"  {name:20} | Cum={cum*100:+.1f}% |
        Sharpe={sh:.3f} | MaxDD={dd*100:.1f}%")

print("\n  Portfolio Performance:")
```

```
perf(port_mv, "Min Variance"); perf(port_ms, "Inverse Vol")

# PART E
last_w   = mv_df.iloc[-1].values
last_vols=
   np.array([garch_fits[a].conditional_volatility.iloc[-1]
   for a in assets])
last_corr= roll_corrs[-1] if roll_corrs else np.eye(3)
last_cov = np.diag(last_vols)@last_corr@np.diag(last_vols)
port_var = np.sqrt(last_w@last_cov@last_w)
var99p   = -2.326*port_var; es99p = -2.665*port_var

np.random.seed(999); n_sims=1000; n_days=126;
   start_val=100.0
mu_port_d = sum(last_w[i]*rets[a].mean() for i,a in
   enumerate(assets))/100
sim_vals  = np.zeros((n_days, n_sims))
for s in range(n_sims):
    val = start_val
    for d in range(n_days):
        cs = np.random.multivariate_normal(np.zeros(3),
           last_corr)
        ar = last_vols*cs/100; pd_r = last_w@ar + mu_port_d
        val = val*(1+pd_r); sim_vals[d,s] = val

p10=np.percentile(sim_vals,10,axis=1);
   p50=np.percentile(sim_vals,50,axis=1)
p90=np.percentile(sim_vals,90,axis=1)

fc_dates = pd.date_range(ret_df.index[-1],
   periods=n_days+1, freq="B")[1:]
fig, ax = plt.subplots(figsize=(16,7))
ax.axhline(100, color="black", lw=0.8, linestyle="--")
ax.fill_between(fc_dates, p10, p90, color="steelblue",
   alpha=0.2, label="80% MC Range")
ax.fill_between(fc_dates,
   np.percentile(sim_vals,25,axis=1),
                np.percentile(sim_vals,75,axis=1),
                   color="steelblue", alpha=0.4,
                   label="50% MC Range")
ax.plot(fc_dates, p50, "b-", lw=2, label="Median Base
   Case")
ax.plot(fc_dates, p10, "r--", lw=1.2, label=f"Bear
   (Rs.{p10[-1]:.1f})")
ax.plot(fc_dates, p90, "g--", lw=1.2, label=f"Bull
   (Rs.{p90[-1]:.1f})")
ax.set_title("6-Month Portfolio Fan Chart (Rs.100 Initial,
   Min-Variance Weights)")
```

```
ax.set_ylabel("Portfolio Value (Rs.)"); ax.legend(fontsize=9)
plt.tight_layout(); plt.show()

print(f"\n  6-Month Scenarios (from Rs.100):")
print(f"    Bear (10th pctile): Rs.{p10[-1]:.2f} ({(p10[-1]-100):.1f}%)")
print(f"    Base (median):      Rs.{p50[-1]:.2f} ({(p50[-1]-100):.1f}%)")
print(f"    Bull (90th pctile): Rs.{p90[-1]:.2f} ({(p90[-1]-100):.1f}%)")
print(f"    P(portfolio > 100): {(sim_vals[-1]>100).mean()*100:.1f}%")

crash_corr = np.ones((3,3))*0.7 + np.eye(3)*0.3
cov_crash  = np.diag(last_vols)@crash_corr@np.diag(last_vols)
shock_btc  = last_vols.copy(); shock_btc[assets.index("BTC")] *= 2
cov_shock  = np.diag(shock_btc)@last_corr@np.diag(shock_btc)

print(f"\n  Stress Scenarios (99% VaR):")
print(f"    Base scenario          : {var99p:.4f}%")
print(f"    COVID-type (3x vol)    : {-2.326*port_var*3:.4f}%")
print(f"    Correlated crash (rho=0.7): {-2.326*np.sqrt(last_w@cov_crash@last_w):.4f}%")
print(f"    BTC vol spike (2x)     : {-2.326*np.sqrt(last_w@cov_shock@last_w):.4f}%")

print(f"""

  INVESTMENT COMMITTEE PRESENTATION SUMMARY
  BTC * MRF * Gold -- Quantitative Portfolio Analysis


  1. PORTFOLIO: Min-variance weights using GARCH cond. vols +
     DCC rolling correlations. Monthly rebalancing.
     Current: BTC={last_w[0]*100:.0f}%, MRF={last_w[1]*100:.0f}%, Gold={last_w[2]*100:.0f}%.

  2. RISK: Port. daily vol={port_var:.4f}% ({port_var*np.sqrt(252):.1f}% annual).
     99% VaR={var99p:.4f}%/day. 99% ES={es99p:.4f}%/day.

```

```
3. GARCH PERSISTENCE:
   BTC={model_results['BTC']['persistence']:.3f},
   MRF={model_results['MRF']['persistence']:.3f},
      Gold={model_results['Gold']['persistence']:.3f}.
   BTC shocks persist longest (half-life
      ~{np.log(0.5)/np.log(model_results['BTC']['persistence']):.0f}d)

4. CROSS-ASSET: VAR({var_fit.k_ar}) shows limited daily
   Granger causality.
   Assets are largely independent risk sources at daily
      frequency.

5. VOL REGIMES: BTC {('HIGH' if
   model_results['BTC']['current_vol_ann']>70 else
   'MODERATE')}
   ({model_results['BTC']['current_vol_ann']:.0f}%
      annual). MRF
      {model_results['MRF']['current_vol_ann']:.0f}%.
   Gold {model_results['Gold']['current_vol_ann']:.0f}%.

6. 6-MONTH FORWARD: Base Rs.{p50[-1]:.0f}
   (+{p50[-1]-100:.1f}%).
   Bull Rs.{p90[-1]:.0f}. Bear Rs.{p10[-1]:.0f}.
   P(portfolio
      gain)={((sim_vals[-1]>100).mean()*100):.0f}%.

7. KEY RISK: Correlated crash (rho=0.7) raises 99% VaR by
   {abs(-2.326*np.sqrt(last_w@cov_crash@last_w)/var99p):.1f}x
      vs base scenario.

8. BTC RISK: BTC allocation is primary risk driver. 2x
   vol
   spike materially increases portfolio VaR.

9. MODEL LIMITS: EWMA DCC approximation; t-distribution
   may still underestimate BTC extreme tails; no VECM.

10. REBALANCING TRIGGER: Monthly or when any GARCH vol
    changes >20% from previous month.

11. RECOMMENDATION: Current min-variance allocation
   appropriate
    for moderate risk mandate. Increase Gold by 5-10% for
    more conservative profile. Implement vol-targeting.

12. EXTENSIONS: (a) Full DCC-GARCH; (b) VECM if
   cointegration
    confirmed; (c) Factor model (dollar, inflation)
       augmentation.
```

```
280
281 """)
```

④ Real World Link

This capstone mirrors the analytical workflow of a multi-asset quantitative research team producing a quarterly risk report. Part E's investment committee summary is the deliverable that reaches portfolio managers and risk committees — it synthesises the full statistical analysis into actionable intelligence. The ability to connect GARCH parameter estimates to investment implications (half-life → position holding period; $\nu \rightarrow$ VaR model choice; DCC → hedge ratio dynamics) is the defining skill of a senior quant in financial risk management.

Module 5 Complete

40 Problems | 5 Easy | 10 Medium | 13 Hard | 12 Exam

Range	Difficulty	Format
M5.001–M5.005	EASY	Guided mini-projects with scaffolding
M5.006–M5.015	MEDIUM	Write from scratch / Find error
M5.016–M5.023	HARD	Advanced write from scratch
M5.024–M5.027	EXAM	Full capstone projects, multi-part
Total	**40 Problems**	All three datasets

Problems	Dataset
M5.001, M5.004, M5.006, M5.009, M5.012, M5.016, M5.022, M5.024	BTC-USD
M5.002, M5.005, M5.007, M5.008, M5.015, M5.017, M5.021, M5.023, M5.025	MRF.NS
M5.003, M5.011, M5.013, M5.018, M5.026	GC=F
M5.010, M5.019, M5.020, M5.027	All three

Complete Workbook Summary

Module	Topic	Problems
M1	Anatomy & Smoothing	40 (M1.001–M1.040)
M2	Stationary Models	40 (M2.001–M2.040)
M3	Forecasting & Model Selection	40 (M3.001–M3.040)
M4	Multivariate & Volatility	40 (M4.001–M4.040)
M5	Integrated Projects	40 (M5.001–M5.040)
Total	**All Modules**	**200 problems**

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